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--- Summary ---

TABLE 1: MOTOR TRADE SALES AT CURRENT PRICES THEN
(RAND IN MILLION)

Month	Year 2018	Year 2019
January	53 044	53 844
February	53 321	52 361
March	58 353	55 646
April	49 633	54 981
May	56 064	56 846
June	55 263	54 249
July	56 409	59 270
August	57 598	58 435
September	54 436	54 467
October	60 406	59 960
November	60 417	57 915
December	51 480	52 718
Total	666 424	670 692

[Source: www.statssa.gov.za/Jan 2020]

Use the information above to answer the following questions.
of units

(4) F

L2

$$\begin{aligned}2.1.3 \text{ \% loss} &= 12\,924,75 - 6\,995,25 \\&= R5\,929,50 \\&= 10075, 1292450, 5929u \\&= 45,88\%\end{aligned}$$

OR

$$\begin{aligned}\text{Percentage loss} &= 10012924,7525, 6995u \\&= 54,12\% \\&= 100\% - 54,12\% \\&= 45,88\% \text{ 1M Subtraction of values}\end{aligned}$$

1S Simplification

1M Dividing correct values

1M Multiply by 100%

OR

1M Dividing correct values 1M M
ultiply by 100%

1S Simplification
1M Subtraction of %

(4) F

L3

2.1.4 B = R8 038,07 – R6 995,25

= R1 042,82 1MA Subtraction 1CA Correct answer (2) F

L2

2.1.5 R765,57 2RT Correct value

(2) F

L2

2.1.6 % increase = 10075,33275,33202,366u■

= 9,998%

= 10% 1RT Correct values

1SF Substitution

1S Simplification

1R Nearest % (4) F

L2

2.2.1 Number of plates 99 RT

2RT Number of plates

(2) F

L2

2.2.2 Fixed expenses = R500 99 RT 2RT Fixed expenses

(2) F

L2

2.2.3 Income = R50 × Number of plates sold 9 M 9 A 1M Multiplication with R50

1A Correct formula (2) F

L2

2.2.4 R0 OR (No Profit) 99 RT 2RT No profit

(2) F

L2 9 R 9 S 99 RT

9 M

9 S

9M 9 CA

9 M 9 C

9 CA

99RT 9C A 9 MA

9RT 9 M

9 M

9 SF 9 S 9 M

9 M

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2.2.5 Loss for 8 plates = Expenses – Income

9RT 9 RT

= 740 – 400 9M

= R340 9A

OR

Expenses = 500 + 8 × 30 = R740 9M

Income = 50 × 8 = R400 9M

Loss = 740 – 400 = R340 9A 1RT R740

1RT R400

1M Subtraction

1A Loss

(From graph allow 3 40±10)

OR
1M for R 740
1M for R400
1M subtraction
1A for R340 exact answer (4) F
L3

[30]

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QUESTION 3: DATA HANDLING (18 marks) AND PROBABILITY (3 marks)

Ques.

Solution Explanation T/L

3.1 Total number of spas

$$= 30\,394 + 13\,856 + 3\,984 + 6\,057 + 46\,282 + 48\,679$$

$$= 149\,252 \text{ 9M 9A}$$

1M Adding correct values

1A Total (2) D

L1

3.2 Mean = ■■■ ■■■

■ 9M

$$= 24\,875,33$$

$$= 24\,875,9 \text{ R CA from 3.1}$$

1M Division

1R Whole number

(2) D

L2

3.3 European spas as a % = ■■ ■■■

$$\frac{\text{■■■}}{\text{■■■}} \times 100 \text{ 9M}$$

$$= 31\% \text{ 9CA 1M Fraction with correct}$$

values and

multiplication by 100

1CA Percentage (2) D

L2

$$3.4 \text{ Range} = 48\,679 - 3\,984$$

$$= 44\,695 \text{ 9S}$$

Number of regions above range = 2 CA 9

1S Calculate range

1CA Number of regions (2) D

L3

$$3.5 \text{ } 30\,394 : 48\,679 = 1 : \text{■■ ■■■}$$

$$\text{■■ ■■■} \text{ 9M 9M}$$

$$= 1 : 1,609 \text{ CA 1M Ratio}$$

1M Fraction

1CA Unit ratio

NPR

(3) D

L3

$$3.6 \text{ Revenue in sub-saharan Africa} = 6,6 - 5,0 \text{ 9M}$$

$$= 1,6 \text{ 9S}$$

Total revenue for spas

$$= 22,9 + 6,6 + 1,6 + 2,8 + 33,3 + 26,5 \text{ 9M}$$

$$= \$93,7 \text{ billion 9CA 1M Subtraction}$$

1S Simplification

1M Addition

1CA Total revenue

Penalise 1 mark if not in

billions (4) D

L3

3.7 1,6; 2,8; 6,6; 22,9; 26,5; 33,3 9M

Median revenue = ■, ■ ■ ■ ■, ■

■ 9M

$$= \$14,75 \text{ billion 9CA CA the value \$1,6 from 3.6}$$

included in the data

1M Arranging in order of
descending or ascending

1M Concept of median

1CA Answer in billions (3) D

L3

3.8 P (Regions with more than 40 000 spas)

$$= \blacksquare$$

$$\blacksquare \times 100 \text{ 9M}$$

$$= 33,33\% \text{ 9CA 1RT Correct numerator and}$$

denominator

1M Multiplication by 100

1CA

Percentage

NPR (3) P

L2

[21]

9RT

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QUESTION 4: FINANCE (12 marks), DATA HANDLING (11 marks) AND
PROBABILITY (3 marks)

Ques.

Study the table and answer the questions that follow:

Table 3: Comparison between urban and rural food prices (selected food items)

Product Rural food

prices

January 2015

(in Rands) Urban food

prices

January 2015

(in Rands) Price

difference

Rand per unit

Full cream Long Life Milk (1 ■) 12,03 12,59 0,56

Loaf of Brown Bread (700 g) 9,57 10,29 0,72

Loaf of White Bread (700 g) 10,31 11,42 1,11
 Maize Meal (5 kg) 32,49 33,73 1,24
 Margarine(500 g) 17,89 21,68 3,79
 Rice (2 kg) 23,62 23,45 -0,17
 Sunflower Oil (750 ml) 14,59 17,25 2,66
 Ceylon/Black Tea (62,5g) 9,89 9,68 -0,21
 White Sugar (2,5 kg) 29,63 26,31 -3,32
 Average A
 [Source: Stats SA, 2015]

4.1 Arrange the rural food prices in descending order.

Study the till

slip below which she received from Sparrow and answer the questions that follow: (*some information is omitted*)

SPARROW SHOP

Roseplace.Tel: 716 1610

Customer: Victoria Prices (R)

Parm Stew F/Cus 1 kg 22,49

Mc Beans (Fresh Greenbeans) 1 kg 36,99#

A/Chix B/P 4 -PCE 0,542 kg @ 53,99 p/kg 29,26

A/Chix B/P 4 -PCE 0,474 kg @ 53,99 p/kg 25,59

Flora regular margarine per kg 2 @ 1 kg 99,98

Sunday Times National 1's 22,00

Bostik Prestik L 100 g 17,99

Koo PCH.Slice SR 410 g 9,99

Valpre Still 1,5 l 13,49

Carrier bag 2 @ 0,50 24 l 1,00

TOTAL FOR 12 ITEMS 278,78

Tendered debit card 478,78

Change cash 200,00

Points earned:

Before This sale: 1639

This sale: 13

Total points : 1652

VAT rate Excluding.

26304/

132568

MODEL Mercedes

Benz Selling

dealer 25A Date 24/03/16

Operation/Part no./Description Quantity/

Time units Unit

Price (R) Unit Amount

Consumables 0100MSH

Engine Oil U 5,50 29 159,50 S

MA271 180 05 09 Filter Oil X 1 212,82 EACH 106,41 S

MA 001 986 19 71/ 1075 Cleaner

Solv X 1 4,99 EACH 2,50 S

MA 000 989 08 07/13 Fluid X 1 75,87 EACH 37,94 S

MA 204 890 2311 TS Wiper Blade X 1 697,13 EACH 348,57 S

MA 201 565 456 03 Battery X 1 2469 ,92 EACH 1234 ,96 S

MA 313 836 0319 Filter Combi X 1 726,16 EACH 363,08 S

Goods

value VAT

RATE Taxable amount

(incl labour) NET TOTAL 3390 ,96

VAT/BTW B

Parts (X) 2093 ,46 14 % 3390 ,96 Rounding 0,09

Labour A

Sublet 0,00

Deposit 0,00 TOTAL 3865 ,60

Goods are supplied subject to ownership vesting in the seller until paid for in full.

--- Full Combined Text ---

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 202 5

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including 2 answer sheet s.

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2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2025)

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INSTRUCTIONS AND INFORMATION

1. Write your FULL NAME and SURNAME on the ANSWER BOOK .
2. The question paper consists of FOUR QUESTIONS. Answer ALL the questions in the ANSWER BOOK .
3. Start EACH question on a NEW page .
4. Number the answers correctly according to the numbering system used in this question paper.
5. Leave ONE line open between two sub -questions , for example between QUESTION 2.1 and QUESTION 2.2.
6. Use the ANSWER SHEETS attached to answer QUESTIONS 2.2.2, 2.2.3 and 3.2.
7. You may use a non -programmable calculator.
8. You may use appropriate mathematical instruments.
9. Show ALL formulae and substitutions in ALL calculations.
10. Round off your final numerical answers to TWO decimal places , unless stated otherwise .
11. Write neatly and legibly.

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(EC/NOVEMBER 2025) MATHEMATICAL LITERACY P1 3
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QUESTION 1

1.1 TABLE 1 presents the deposit slip details for funds to be credited to Cynthia Francis 's account .

TABLE 1: CYNTHIA FRANCIS'S DEPOSIT SLIP

Deposit Slip
CFO BANK
Credit Cynthia Francis
15 December 2024
Teller's stamp Paid by: Cynthia Francis R c
Signature : C Francis Notes 260 00
Cell no: 061 320 4988 Coins 16 70
Subtotal 276 70
Authorised by Drawer's name Bank Branch
Authority no. 1. A Troyd BBC Jamestown 340 00
2. Z Pasi DBE Bloemfontein 480 80
Total
Account no. 2 4 5 7 1 6 4 8 3 4

1.1.1 Write down Cynthia's account number . (2)

1.1.2 Determine how many transfers were from other banks . (2)

1.1.3 Write down the total amount of cash that she deposited. (2)

1.1.4 Calculate the total of the amount deposited . (2)

1.2 Zizo, a soccer fan, collected and analysed 2024/2025 goals scored for the Premium Soccer League. The goals scored for the league are indicated below.

TABLE 2 : 2024/2025 PREMIER SOCCER LEAGUE GOALS SCORED

TEAMS SFC ORP SUN AMA TSG KAC CHI MGE CTC

GOALS

SCORED

73

60

46

35

35

32

31

30

27

Use the information above to answer the questions that follow.

1.2.1 Choose the correct word to make the statement TRUE.

The above data is numerical / categorical. (2)

1.2.2 Write down the highest number of goals scored in the league. (2)

1.2.3 List the teams that have the same number of goals scored. (2)

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4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2025)

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1.3 Choose a definition from COLUMN B that matches the term in COLUMN A. Write only the letter (A–F) next to the question numbers (1.3.1 to 1.3.4) in the ANSWER BOOK , for example 1.3.5 G .

Column A Column B

1.3.1 Questionnaire A the likelihood of something happening or not happening

1.3.2 Balance B a record of income and expenditure

1.3.3 Probability C instrument used to collect data

1.3.4 UIF D payment made into the bank account

E the difference between the deb its and credits

F unemployment insurance fund

(4 x 2) (8)

[22]

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QUESTION 2

2.1 Rowe has been driving a demo car for the past four years. Over these four years he managed to save 16% as a deposit of the price of the brand new car he wants to buy. He is interested in buying a 2025 Cherry Tiggo 7 Pro Max.

Rowe visited the dealer to purchase the car. The price of the car is R499 900,00.

NOTE : Focus will be on the price of the car only (no other initial costs).

Use the above information to answer the following questions.

2.1.1 Calculate the amount Rowe saved for the deposit. (2)

2.1.2 Rowe secured a loan for the outstanding balance from the bank, agreeing to repay it over 36 months with a 9, 5% annual simple interest rate.

Calculate how much Rowe will pay if it takes him exactly 36 months to pay the loan. (5)

2.1.3 Determine the monthly instalment Rowe will pay on the loan . (2)

2.1.4 In the showroom there were different colours of the car Rowe wants to buy (2 red, 3 silver, 4 grey and 5 black). Calculate the probability , as a decimal , that Rowe will choose a silver car . (3)

2.2 Waterfront High School wants to renovate their netball court. The school asked builders from the community to volunteer and assist with the renovations. The school estimated that it would take 3 builders 12 days to finish renovations, or it would take 12 builders 3 days to finish renovations. The table below displays this information.

TABLE 3: NUMBER OF BUILDERS AND DAYS TO RENOVATE THE
NETBALL COURT
Number of builders 3 4 6 9 12
Number of days 12 3

Use the above information to answer the following questions.

2.2.1 Derive a mathematical formula that represents the information given in the table . (2)

2.2.2 Complete the table above by using the ANSWER SHEET for QUESTION 2.2.2 provided . (3)

2.2.3 Use ANSWER SHEET QUESTION 2.2.3 provided to draw a graph to represent the data in TABLE 3. (4)

2.2.4 Choose the correct word to make the statement false.

The relationship represented by the graph in QUESTION 2.2.3 is direct / indirect proportion . (2)

2.3 Joyce's salary is R20 600,00 per month. Last year her monthly salary was R20 000,00.

Calculate the inflation rate. (4)
[27]

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6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2025)
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QUESTION 3

3.1 The pie chart below shows the percentage of bachelor passes obtained by candidates in Cluster B Districts. The total number of bachelor passes in Cluster B Districts is 20 910.

3.1.1 Determine the percentage of bachelor passes obtained by the Sarah Baartman District. (3)

3.1.2 Calculate the number of candidates with bachelor passes in Buffalo City. (3)

3.1.3 Determine the median percentage of bachelors passes. (3)

3.1.4 Write down the district that has the least percentage of bachelor passes . (2)

3.1.5 Calculate the mean percentage of bachelor passes in Cluster B. (3)

3.1.6 Determine the probability of randomly selecting a district with a bachelor pass of less than 10%. Write your answer as a simplified fraction. (3)

3.2 A clinic recorded the distribution of a few antibiotics to a sample group of learners at a school to see whether they contracted TB . The sample group was made up of nine classes. The results are shown in the table below.

TABLE 4 : THE RESULTS OF LEARNERS WHO CONTRACTED TB

Classes A B C D E F G H I

Number of antibiotics 35 30 10 50 45 35 20 55 10

Number of learners contracted TB 40 45 50 5 20 25 50 10 55

Draw a scatter plot to illustrate the data in the table above. Use the ANSWER SHEET provided for QUESTION 3.2. (6)

[23] Amathole

East 20%

Amathole West

8%

Buffalo

City 26%

Chris Hani West

12%Nelson

Mandela

27%Sarah Baartman

Amathole East Amathole West Buffalo City

Chris Hani West Nelson Mandela Sarah BaartmanBACHELOR PASS PER DISTRICT

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QUESTION 4

4.1 The invoice given below lists cleaning materials bought by Lily. Study the invoice and answer the questions that follow.

INCOME AND EXPENDITURE

DESCRIPTION QUANTITY LITRES PRICE PER LITRE COST

1. Water -based general purpose

cleaner 2 25 R99, 00 R4 950

2. Pine fragrance disinfectant

cleaner 2 25 R95, 00 R4 750

Sub Total ...

VAT (15%) R1 455

Total Due ...

4.1.1 Calculate the total cost for ALL the material bought by Lily , including VAT . (4)

4.1.2 Lily states that the profit she makes from selling ONE 25 litre (25 ■) Pine fragrance disinfectant cleaner is R450,00, if she sells a litre at R135, 00.

Verify , showing ALL calculations, if her statement is VALID. (6)

4.1.3 The graph below shows the cost and selling price of all-purpose cleaner material per litre. Study the table and answer the questions that follow.

(a) Give the value(s) of the point marked (W) on the graph and explain the meaning of that point. (2)

(b) Write down the formula for Lily's expenses if she must pay rent of R2 00.

Cost = ... + ... (2) 05001000150020002500300035004000

0 5 10 15 20 25 30 Cost/Selling price (Rands)

Number of litres COST AND SELLING PRICE OF ALL -PURPOSE CLEANER

WINCOME

EXPENSES

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8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2025)

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4.2 A South African company imports goods worth €45 000 from Europe.

The exchange rate is R1 = € 0,049669

Calculate the cost of the goods in South African Rand. (3)

4.3 Below are the March Test results for nine Grade 11 Mathematical Literacy learners

The test was out of 75 marks.

15 60 75 67 68 73 60
75 58

The teacher claimed that the range of the class is 80%. Show by means of calculations whether the teacher's claim is VALID or not. (5)

4.4 The City of FY2024/25 split the residential prepaid tariffs into Prepaid Low (indigent)

and Prepaid High. Electricity tariffs for Prepaid Low are shown in TABLE 5.

TABLE 5: RESIDENTIAL PREPAID LOW INDIGENT (FY 2024/25)

SIZE TARIFF (kWh/R)

(Excluding VAT)

Block 1 0 – 350 kWh 2,216

Block 2 ■ 350 – 500 kWh 2,711

Block 3 ■ 500 kWh 3,226

Use TABLE 5 above to answer the question that follows.

Calculate how much a resident in this city pays a month, including VAT, when they use 374 kWh of electricity. (6)

[28]

TOTAL: 100

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ANSWER SHEET

NAME OF SCHOOL:.....CLASS:

NAME OF LEARNER:

QUESTION 2.2.2

NO. OF BUILDERS 3 4 6 7 12

NO. OF DAYS 12 3

QUESTION 2.2.3

02468101214

0 2 4 6 8 10 12 14.....

.....
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ANSWER SHEET

NAME OF SCHOOL:..... CLASS:

NAME OF LEARNER:

QUESTION 3.2

CLASSES A B C D E F G H I

Number of antibiotics 35 30 10 50 45 35 20 55 10

Number of learners contracted TB 40 45 50 5 20 25 50 10 55

51525354555

10 15 20 25 30 35 40 45 50 55 60NUMBER OF LEARNERS CONTRACTED TB

NUMBER OF ANTIBIOTICSSCATTER PLOT SHOWING ANTIBIOTICS AND NUMBER OF LEARNERS CONTRACTED TB

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NATIONAL

SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2022

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 12 pages , including two answer sheets.

2 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2022)

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. 2.1 Use ANSWER SHEET 1 for QUESTION 4.1.3 .

2.2 Use ANSWER SHEET 2 for QUESTIONS 4.3.1 .

2.3 Write your NAME and GRADE in the spaces provided on ANSWER SHEET 1 for QUESTION 4.1.3 AND on ANSWER SHEET 2 for QUESTION 4.3.1 .
Hand in the ANSWER SHEETS with your ANSWER BOOK.

3. Number the answers correctly according to the numbering system used in this question paper.

4. Diagrams are NOT necessarily drawn to scale , unless stated otherwise .

5. Round off ALL final answers according to the context used, unless stated otherwise.

6. Indicate units of measurement, where applicable.

7. Start EACH question on a NEW page.

8. Show ALL calculations clearly.

9. Write neatly and legibly.

(EC/ NOVEMBER 2022) MATHEMATICAL LITERACY P1 3

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QUESTION 1

1.1 Study Ms Jane Bukirwa's payslip for the month of October 2021 shown below and answer the questions that follow.

GM's Car Washing Co. Ltd.

8 Down Street,

Komani,

5319

Tel: _____

Cellphone no.: _____

[www.yorkshire.co.uk]

Payslip for the month of October 2021

Employee 's name: Jane Bukirwa

Gender: Female

Occupation: Clerk/Cashier Paid days: 30
 Leave without pay: 1
 Pay slip no.: 10/2021
 Earnings Amount Deductions Amount
 Basic salary R5 000 Salary advance R650
 Overtime hours 45 PAYE R774
 Overtime rate R25 UIF: 1% of Total
 Earnings
 R61,25
 Overtime payment R1 125 Other deductions -
 Total Payment R6 125 Total deductions ----
 Net Pay R4 639,75

1.1.1 Write down Ms Jane Bukirwa's occupation. (2)

1.1.2 Calculate her total deductions. (2)

1.1.3 Show by means of calculations how the overtime payment amount of R1 125 was determined. (2)

1.1.4 Define the term ' net pay ' according to the above context . (2)

1.1.5 Calculate the amount to be added to Jane Bukirwa's basic salary , if the basic salary is increased by 7,5 % for November 2021. (3)

4 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2022)

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1.2 Grade 11 Mathematical Literacy learners wrote a test marked out 50 marks. Their results are displayed on the graph below showing the distribution of marks.

Use the information displayed on the above graph to answer the questions that follow.

1.2.1 Name the type of graph used to display the data of results. (2)

1.2.2 Determine the number of learners that wrote the test. (2)

1.2.3 How many learners scored within the modal class? (2)

1.2.4 Calculate the marks out of 50 for a learner that obtained 68% in the test. (2)
 [19]

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QUESTION 2

2.1 Aunt Louis, an educator, compares the pension fund estimations for the year 202 1 and 202 2. She wants to resign at the end of 202 2. Study the information below and answer the questions that follow s.

Normal retirement estimated 2022 2021

Monthly pension (annuity) payable

R26 383 R25 591

Lump sum (gratuity) payable R919 363 R884 198

Discharge monthly pension payable R23 248 R22 083

Discharge lump sum (gratuity) payable R799 313 R747 619

Monthly pension (annuity) payable to spouse(s) R13 692 R13 295

Lump sum (gratuity) payable to beneficiaries R2 592 409 R2 470 783

Resignation benefit R2 836 836 -----

[Adapted from www.gepf.co.za]

2.1.1 Write down the value of the lump sum payable to beneficiaries in 2021. (2)

2.1.2 The resignation benefit for the year 2021 is R166 417 less than 2022 benefit.

Calculate the resignation amount for the year 2021. (3)

2.1.3 Calculate the annual amount for the year 2022 that Aunt Louis will receive for the monthly pension. Round off your answer to the nearest R100. (3)

2.1.4 Aunt Louis claimed that the lump sum (gratuity) payable for 2022 is R35 615 more than the lump sum (gratuity) payable for 2021. Verify the statement. (3)

2.1.5 Give ONE advantage of an employee contributing towards the Pension Fund. (2)

6 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2022)

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2.2 John Matu sells Samsung cellphones. The price list given below shows the cost price and profit for different types of cellphones in the year 2021. Study the price list and answer the questions that follow. All prices include VAT 15%.

Samsung Galaxy A31

Cost Price: R3 299,25

Profit: R1 099,75

Samsung Galaxy A72

Cost Price: R7 499,25

Profit: R2 499,75

Samsung Galaxy S10

Cost Price: R11 099,25

Profit: R3 699,75

Samsung Galaxy S21 Ultra

Cost Price: R21 749,25

Profit: R7 249,75

[Source: www.samsung.com]

2.2.1 Determine the selling price of the Samsung Galaxy A72. (2)

2.2.2 Determine the ratio (in simplest form) of total profit to the total income from sale of the four cellphones. (4)

2.2.3 John Matu ran out of the cheaper phones. He placed an order for the cellphones A31 and A72 in the ratio: A31 : A72 = 2 : 1.

A total of 15 cellphones were ordered.

Verify with the necessary calculations that the total cost price for the order was R70 488,75. (5)

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2.2.4 The cost price of Samsung A31 in the year 2020 was R3 399,75. It is costing less this year due to new cellphones on the market.

Calculate the percentage decrease on the cost of the A31. Give your answer to the nearest percentage.

You may use the following formula:

Percentage decrease = $\frac{\text{new price} - \text{old price}}{\text{old price}} \times 100\%$ (4)

2.3 Mr John Matu received the statement indicating water tariffs for the month of March 2021. Study the water tariffs table below from DJB Municipality.

DJB Municipality

Mr J. Matu : March 2021 statement

Code Water tariffs (per k ■)

(Excluding VAT)

WA0101 6,110 k ■ @ R16,992900

WA0102 4,073 k ■ @ R17,326130

WA0103 2,817 k ■ @ R24,06370

Amount paid R ---

Use the information and the table above to answer the following question .

Mr J. Matu asked his two sons to calculate the amount (including VAT = 15%) he has to pay.

- Son A got R2 63,48
- Son B got R2 58,48

John claimed Son B was correct. Verify showing ALL your calculations whether the statement is correct. (5)
[33]

8 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2022)

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QUESTION 3

Motor trade sales in the months (January –December) for the years 2018 and 2019 at current prices then, are shown in the TABLE 1 below.

TABLE 1: MOTOR TRADE SALES AT CURRENT PRICES THEN
(RAND IN MILLION)

Month	Year 2018	Year 2019
January	53 044	53 844
February	53 321	52 361
March	58 353	55 646
April	49 633	54 981
May	56 064	56 846
June	55 263	54 249
July	56 409	59 270
August	57 598	58 435
September	54 436	54 467
October	60 406	59 960
November	60 417	57 915
December	51 480	52 718
Total	666 424	670 692

[Source: www.statssa.gov.za/Jan 2020]

Use the information above to answer the following questions.

3.1 Write down the month and the year when trade sales were the lowest of the two years. (2)

3.2 Determine the range for the amount of trade sales for the months in the year 2019. (3)

- 3.3 Write down the total amount of the trade sales for the year 2018 in words. (2)
- 3.4 Calculate the mean trade sales for the year 2018. (3)
- 3.5 Determine the median value of the trade sales for the months (Jan –Dec) in the year 2019. (3)
- 3.6 Express the amount of trade sales of October 2018 as a percentage of the total trade sales for the year 2018. (3)
- 3.7 Determine the probability (as a fraction) of randomly selecting a month that shows a decline in the amount of trade sales from year 2018 to 2019. (3)

(EC/ NOVEMBER 2022) MATHEMATICAL LITERACY P1 9
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QUESTION 4

4.1 Ms Jena started a shoe repair business to repair old shoes , restore their quality and make them user friendly. Below are the weekly costs and charges involved :

- Rent for the room – R250
- Salaries – R400
- Materials required per pair of shoes on average costed at R25
- On average each customer is charged R75 per pair of shoes
- Income = $R75 \times n$, (where n represents the number of pair s of shoes)
- Expenses = $R650 + R25 \times n$, (where n represents the number of pairs of shoes)

[Source: <https://howtostatanllc.com/Shoe repair>]

Use the above information to answer the following questions.

- 4.1.1 Write down the weekly total fixed costs that Ms Jena pays per week. (2)
- 4.1.2 The TABLE 2 below shows the income and expenses from the business.

TABLE 2: INCOME AND EXPENSES FOR THE SHOE REPAIR BUSINESS

Number of pair of shoes (n)	0	5	10	15	18	20
Expenses (R)	650	775	900	1 025	1 100	A
Income (R)	0	375	750	1 125	1 350	1 500

- (a) Calculate missing value A. (2)
- (b) Ms Jena has learnt that starting a business is not easy. It takes time to get your business known. Her records show that:
- In week 1 – She repaired 5 pairs of shoes
 - In week 2 – She repaired 10 pairs of shoes
 - In week 3 – She repaired 20 pairs of shoes

Use the information above and TABLE 2 to show by means of calculations that Ms Jena made a loss of R200 in the 3 weeks. (5)

4.1.3 A line graph showing the income from the repair of shoes has been drawn on ANSWER SHEET 1. Use the same grid on ANSWER SHEET 1 to draw another line graph showing the expenses of the repair of shoes. (3)

4.1.4 Use the graph to determine the number of pair of shoes at the break-even point. (2)

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4.2 Ms Jane's brother who lives in Japan sent her money to assist her in her business. He sent Japanese yen (¥) 25 000. Determine the amount in Rands Ms Jane received.

You may use the exchange rate at the time: 1Yen (¥) = R0,135 2364 (3)

4.3 A Grade 11B Mathematical Literacy class at Nayani Secondary School participated in a research project where their gender and weight (kg) were recorded. The raw data is indicated in TABLE 3 below:

TABLE 3: GENDER AND WEIGHTS FOR GR ADE 11B

Gender Weights in kg

Female 58 67 73 68 80 65 90 78 108 59 67 79 75

Male 61 57 70 72 66 81 93 77 79 83 75 82

Use the above information to answer the following questions .

4.3.1 Complete the frequency table that the teacher left incomplete . Use ANSWER SHEET 2.

Weight

Category Female Male

Tally Frequency Tally Frequency

50–59 // 2 / 1

60–69 //// 4

70–79 //// 4

80–89 / 1

90–99 / 1

100–109 / 1

(5)

Use the above information in TABLE 3 and your completed ANSWER SHEET 2 to answer the questions that follow .

4.3.2 In which category do most of the female s fall? (2)

4.3.3 Calculate the number of m ales who weigh s 70 kilograms (kg) or more but less than 90 kg. (2)

4.3.4 Determine the probability (as a percentage) of randomly selecting a learner weighing 70 kg or more from the class. (3)

[29]

TOTAL : 100

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ANSWER SHEET 1

QUESTION 4.1.3

NAME OF SCHOOL: _____

GR 11:

NAME OF LEARNER: _____

Income and Expenses (R)
Number of pairs of shoes (n)GRAPH FOR INCOME AND EXPENSES SHOE REPAIR
BUSINESS
INCOME
12 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2022)
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ANSWER SHEET 2

QUESTION 4.3.1

NAME OF SCHOOL: _____

GR 11: _____

NAME OF LEARNER: _____

Complete the frequency table that the teacher left incomplete:
Use the table in the ANSWER SHEET 2 .

Weight Category Female Male
Tally Frequency Tally Frequency
50–59 // 2 // 2
60–69 //// 4
70–79 //// 4
80–89 / 1
90–99 / 1
100–109 / 1

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P1
EXEMPLAR

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages, including an answer sheet.

Imlit1

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)
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INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. 2.1 Use the ANSWER SHEET for QUESTION 4.3.2.

2.2 Write your NAME and GRADE in the spaces provided on the ANSWER SHEET for QUESTION 4.3.2.
Hand in the ANSWER SHEET with your ANSWER BOOK.

3. Number the answers correctly according to the numbering system used in this question paper.
4. Maps and diagrams are not necessarily drawn to scale, unless stated otherwise.
5. Round off ALL final answers according to the context used, unless stated otherwise.
6. Indicate units of measurement, where applicable.
7. Start EACH question on a NEW page.
8. You may use an approved calculator (non-programmable and non-geographical), unless stated otherwise.
9. Show ALL calculations clearly.
10. Write neatly and legibly.

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 3
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QUESTION 1

1.1 Moses earns a gross salary of R10 500 per month. The basic monthly expenses used from the income are as follows:

Housing expense 21%; Food 36%, Transport 10%; Cellphone bills 1,9 %. The rest is for savings.

Use the above information to answer the following questions.

1.1.1 Calculate the total gross salary per annum. (2)

1.1.2 Determine the amount of the monthly food expense. (2)

1.1.3 Write down the ratio of the housing percentage to the food percentage in the simplest form. (2)

1.1.4 Calculate Moses 's percentage for savings from his salary. (3)

1.2 Thando, a Mathematical Literacy teacher, collected and analysed test results of his class. The test was marked out of a total mark of 50. The results of the learners are indicated below:

38 21 12 18 41
28 24 34 10 35

Use the above information to answer the questions that follow.

1.2.1 Is the above data primary or secondary? (2)

1.2.2 Write down the highest mark obtained in the test. (2)

1.2.3 Explain the meaning of the term ' median '. (2)

1.2.4 Write down the mark out of 50 for a learner who achieved 70% in the test. (2)

1.2.5 Write down the number of learners who failed the test, if the pass mark is 20 out of 50. (2)

1.3 A shopkeeper bought a dress for R750 and sold it, making a loss of R50.

1.3.1 Explain the meaning of the term 'loss' in this context. (2)

1.3.2 Calculate the percentage loss made on the sale. (2)
[23]

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 2

2.1 Mrs Rogue invested money monthly from 2016 to 2020 at Company A. She terminated her investment and received the statement below from Old Mutual:

Contract number Contracting
party Name of
Investment plan Start date
of contract Last
premium
due date

17801249
R. Rogue Smart MAX
Focused
Education Plan/1
01/07/2016
30/06/2031
Fund value as at 23 March 2020
Investment fund Unit price (cents) Number of
units Fund value
Allan Gray Balanced Fund C Class 8 266,470 A R8 038,07
Total R8 038,07

Difference between fund value and termination value
Fund value R8 038,07
Reduction fees and transaction charges B
Termination value R6 995,25
Withdrawals up to 23 March 2020 R765,57

Expected premium details as at 23 March 2020
Current
premium
Annual
Increase
rate Next
premium
increase New premium Investment
fund(s) Premium
Split
R332,75
monthly ... 01/07/2020 R366,02 Allan Gray
Balanced Fund
C Class 100%
Total contributions since contract start date
Total premiums paid R12 924,75

2.1.1 Write down the name of the investment plan that Mrs Rogue has. (2)

2.1.2 Calculate the value of A, the number of units earned from the given fund value.
(4)

2.1.3 Show by means of calculations that Mrs Rogue lost 45,88% of the total premiums she

contributed towards the fund by terminating her investment before the last premium. (4)

2.1.4 Calculate the value (in Rands) of B, the reduction fees and transaction charges. (2)

2.1.5 Write down the withdrawal value as at 23 March 2020. (2)

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 5

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2.1.6 Calculate the percentage increase on the monthly premium. Give your final answer to the nearest percentage.

You may use the following formula:

Percentage increase = $\frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100\%$ (4)

2.2 Alice sells food from a kiosk that she rents in a township. The expenses associated with

ingredients and labour for a plate amount to R30.

The formula for expense is: Expenses = R500 + R30 × Number of plates

She sells food per plate. The graph below shows her expenses and income for a month.

Use the above information and the graph to answer the questions that follow.

2.2.1 Write down the independent variable in the above context. (2)

2.2.2 Determine the amount Alice pays as fixed expenses. (2)

2.2.3 Write down the formula for the income if a plate is sold at R50, in the form of:
Income = ... (2)

2.2.4 Determine the profit at the break-even point. (2)

2.2.5 Determine the loss when 8 plates are sold. (4)

[30] 020040060080010001200140016001800

0 5 10 15 20 25 30 35 Amount in Rands(R)

Number of plates ALICE'S EXPENSES AND INCOME FROM SALES

Expenses

Income

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 3

The Global Wellness Institute did research about the amount of spa revenue generated in different regions of the world in 2017 .

TABLE 1: NUMBER OF SPAS, REGIONS AND THE REVENUE GENERATED

Regions Names of Regions Number of Spas Revenue in Billion dollars (\$b)

P North America 30 394 22,9

Q Latin America-Caribbean 13 856 6,6

R Sub-Saharan Africa 3 984 ...

S Middle East – North Africa 6 057 2,8

T Europe 46 282 33,3

V Asia – Pacific 48 679 26,5

TOTAL

[Source: globalwellnessinstitute.org]

Use the information above to answer the questions that follow.

- 3.1 Calculate the total number of spas used for the research in 2017. (2)
- 3.2 Calculate the mean number of spas used. Give your final answer to the nearest whole number. (2)
- 3.3 Express the number of spas in Europe as a percentage of the total number of spas. (2)
- 3.4 Determine the number of regions that lies above the range number of spas. (2)
- 3.5 Write down the unit ratio of spas in regions P and V. (3)
- 3.6 Calculate the total revenue in billion dollars for the spas if sub-Saharan Africa's revenue is \$5,0 billion less than Latin America-Caribbean's revenue. (4)
- 3.7 Calculate the median revenue of the regions. (3)
- 3.8 Determine the probability (as a percentage) of randomly selecting a region with more than 40 000 spas. (3)

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QUESTION 4

4.1 GOOD BOYS CAR WASHING BAY is in an area where water is charged according to the water tariffs structure shown in TABLE 2 below.

TABLE 2: WATER TARIFF STRUCTURE

Block Usage in kilolitre (k ■)	Normal Charge per kilolitre (k■) (Excluding VAT)
1 0 – 6	R0,00
2 +6 – 15	R9,35
3 +15 – 30	R11,16
4 +30 – 45	R12,53
5 +45 – 60	R13,98
6 60+	R15,34

NOTE: VAT is Value Added Tax. The VAT rate is 15%.

Use TABLE 2 above to answer the question that follow.

Calculate how much GOOD BOYS CAR WASHING BAY pays a month including VAT when they use 25 k ■ of water and give a reason why a step up (increasing block rate) system of water tariffs is used to charge water consumption other than a flat single rate. (6)

4.2 Andile received a donation from his brother who works in Rwanda. The donation was 745 614,04 Rwanda Francs (RWF). The bank deducted 10% for bank charges. Andile states that he will receive R12 750. Verify the statement with the necessary calculations.

Use the exchange rate $R\ 0,019 = 1\ RWF$ (6)

4.3 Ibanda High School recorded absentees in the Grade 11 Mathematical Literacy class during the first week when schools reopened after the COVID-19 pandemic break. The data is shown in the table below.

Monday	Tuesday	Wednesday	Thursday	Friday
Girls 4	6	8	7	10

Boys 5 3 8 9 7

Use the above information to answer the questions that follow.

4.3.1 Determine the probability that a learner chosen at random was absent on Wednesday. (3)

4.3.2 Complete the double bar graph on the ANSWER SHEET provided by plotting the missing bars for the number of boys and girls absent. (5)

8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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4.4 A Grade 11 Mathematical Literacy class at Ibanda High wrote an examination marked out of 100 marks. The results arranged in ascending order are shown below.

23 41 42 50 50 51 54 55 56 57
60 61 65 66 66 67 68 69 70 70
70 72 C 74 76 79 82 85 86 88

Use the above information to answer the questions that follow.

4.4.1 The mean of the above data is equal to 64,2. A learner calculated that the value of C in the above data is 74. Verify , with the necessary calculations, whether the answer is valid. (4)

4.4.2 The frequency table of the above data is shown below.

Class interval Frequency

20–29 1

30–39 D

40–49 2

50–59 7

60–69 8

70–80 8

80–89 4

Determine the value of D and give a reason for your answer. (2)

[26]

TOTAL: 100

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ANSWER SHEET

QUESTION 4.3.2

NAME OF LEARNER:

GRADE 11:

024681012

Monday Tuesday Wednesday Thursday Friday Number of absent learners

Days of the week Number of absent learners during the week

Girls Boys

NATIONAL

SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P1
MARKING GUIDELINE
EXEMPLAR

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

AO Answer only

NPR No penalty for rounding

This marking guideline consists of 8 pages.

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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MARKING GUIDELINES

NOTE:

- x If a candidate answers a question TWICE, only mark the FIRST attempt.
- x If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version)
- x Consistent accuracy (CA) applies in ALL aspects of the marking guidelines, however it stops at the second calculation error.
- x If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra incorrect item presented.

LET WEL:

- x As ■ kandidaat ■ vraag TWEE keer beantwoord, merk slegs die EERSTE poging.
- x As ■ kandidaat ■ antwoord van ■ vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.
- x Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyn toegepas, maar dit hou by die tweede berekeningsfout op.
- x Wanneer ■ kandidaat aflesings vanaf ■ grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra verkeerde item.

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QUESTION 1 [23 marks]

Ques. Solution Explanation T&L

1.1.1 Annual gross salary = $R10\,500 \times 12$
= R126 000

1M Multiply by 12

1A Gross per annum (2) F

L1

1.1.2 Monthly food expense = R10 500 × 36%
 = R3 780 1M % Calculation
 1CA Amount (2) F
 L1

1.1.3 Housing % : Food %
 = 21% : 36%
 = 7 : 12 9CA
 1M Correct values and order 1C
 A Simplest form (2) F
 L1

1.1.4 Savings % = 100% – (21% + 36% + 10% + 1,9%)
 = 100% – 68,9%
 = 31,1% 1M Adding correct values
 1M Subtracting from 100
 1CA Percentage (3) F
 L1

1.2.1 Primary data
 2A Correct data type
 (2) D
 L1

1.2.2 41 99 RT
 2RT Highest mark
 (2) D
 L1

1.2.3 Median is the middle value of a set of data which is arranged
 from small to big. 2A Explanation
 (2) D
 L1

1.2.4 35
 2A Correct mark
 (2) D
 L1

1.2.5 3 2RT No. of learners failed
 (2) D
 L1

1.3.1 Loss is when the cost is more than the income. 99 A
 OR
 Loss incurred when selling price is less than cost price of an
 item. 99 A 2A Correct explanation
 (2) F
 L1

1.3.2 % loss = ■■■
 ■■■ × 100% 9M
 = 6,67% 9 CA 1M Fraction multiplied by
 100%
 1CA Percentage
 NPR (2) F
 L1
 [23]
 9 M
 9 A
 9M

9M
99A
99 RT 9M
9CA
99 A 9M

9CA
99A

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 2: FINANCE [30 marks]

Ques. Solution Explanation Topic

/Level

2.1.1 SmartMAX Focussed Education Plan 1 2A Correct investment plan
(2) F

L1

2.1.2 Number of units =

100470, 8266

= R82,6647

= 6647,8207, 8038

= 97,23703104 1C Converted to Rands

1CA Value

1M Division

1CA No. of units

(4) F

L2

2.1.3 % loss = 12 924,75 – 6 995,25

= R5 929,50

= 10075, 1292450, 5929u

= 45,88%

OR

Percentage loss = 10012924,7525, 6995u

= 54,12%

= 100% – 54,12%

= 45,88% 1M Subtraction of values

1S Simplification

1M Dividing correct values

1M Multiply by 100%

OR

1M Dividing correct values 1M M

ultiply by 100%

1S Simplification

1M Subtraction of %

(4) F

L3

2.1.4 B = R8 038,07 – R6 995,25

= R1 042,82 1MA Subtraction 1CA Correct answer (2) F

L2

2.1.5 R765,57 2RT Correct value

(2) F

L2

2.1.6 % increase = 10075,33275,33202,366u■

= 9,998%

= 10% 1RT Correct values

1SF Substitution
1S Simplification
1R Nearest % (4) F
L2

2.2.1 Number of plates 99 RT
2RT Number of plates
(2) F
L2

2.2.2 Fixed expenses = R500 99 RT 2RT Fixed expenses
(2) F
L2

2.2.3 Income = R50 × Number of plates sold 9 M 9 A 1M Multiplication with R50
1A Correct formula (2) F
L2

2.2.4 R0 OR (No Profit) 99 RT 2RT No profit
(2) F

L2 9 R 9 S 99 RT
9 M
9 S
9M 9 CA
9 M 9 C
9 CA
99RT 9C A 9 MA
9RT 9 M
9 M
9 SF 9 S 9 M
9 M

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2.2.5 Loss for 8 plates = Expenses – Income
9RT 9 RT
= 740 – 400 9M
= R340 9A

OR

Expenses = $500 + 8 \times 30 = R740$ 9M
Income = $50 \times 8 = R400$ 9M
Loss = $740 - 400 = R340$ 9A 1RT R740
1RT R400
1M Subtraction
1A Loss
(From graph allow $3\ 40 \pm 10$)
OR
1M for R 740
1M for R400
1M subtraction
1A for R340 exact answer (4) F
L3

[30]

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)
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QUESTION 3: DATA HANDLING (18 marks) AND PROBABILITY (3 marks)
Ques. Solution Explanation T/L
3.1 Total number of spas

$= 30\,394 + 13\,856 + 3\,984 + 6\,057 + 46\,282 + 48\,679$
 $= 149\,252\,9\text{M}$ 9A
 1M Adding correct values
 1A Total (2) D
 L1

3.2 Mean = ■■■ ■■■
 ■ 9M
 $= 24\,875,33$
 $= 24\,875\,9\text{R}$ CA from 3.1
 1M Division
 1R Whole number
 (2) D
 L2

3.3 European spas as a % = ■■ ■■■
 ■■■ ■■■ $\times 100$ 9M
 $= 31\%$ 9CA 1M Fraction with correct
 values and
 multiplication by 100
 1CA Percentage (2) D
 L2

3.4 Range = $48\,679 - 3\,984$
 $= 44\,695\,9\text{S}$
 Number of regions above range = 2 CA 9
 1S Calculate range
 1CA Number of regions (2) D
 L3

3.5 $30\,394 : 48\,679 = 1 : \text{■■} \text{■■}$
 ■■ ■■■ 9M 9M
 $= 1 : 1,60\,9$ CA 1M Ratio
 1M Fraction
 1CA Unit ratio
 NPR (3) D
 L3

3.6 Revenue in sub-saharan Africa = $6,6 - 5,0$ 9M
 $= 1,6\,9\text{S}$
 Total revenue for spas
 $= 22,9 + 6,6 + 1,6 + 2,8 + 33,3 + 26,5$ 9M
 $= \$93,7$ billion 9CA 1M Subtraction
 1S Simplification
 1M Addition
 1CA Total revenue
 Penalise 1 mark if not in
 billions (4) D
 L3

3.7 1,6; 2,8; 6,6; 22,9; 26,5; 33,3 9M

Median revenue = ■, ■ ■■■, ■
 ■ 9M
 $= \$14,75$ billion 9CA CA the value \$1,6 from 3.6

included in the data
 1M Arranging in order of
 descending or ascending
 1M Concept of median
 1CA Answer in billions (3) D
 L3

3.8 P (Regions with more than 40 000 spas)

$\frac{\text{■}}{\text{■}} \times 100$ 9M
 $= 33,33\%$ 9CA 1RT Correct numerator and
 denominator
 1M Multiplication by 100
 1CA
 Percentage
 NPR (3) P
 L2

[21]
 9RT
 (EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 7
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 QUESTION 4: FINANCE (12 marks), DATA HANDLING (11 marks) AND
 PROBABILITY (3 marks)

Ques. Solution Explanation T/L

4.1 Cost =
 Kilolitre Cost
 $6 \times 0 = 0$ 9M
 $9 \times 9,35 = R84,15$
 $10 \times 11,16 = R111,6$
 Total = 25 litres $84,15 + 111,60 = R195,75$
 $R195 \times 115\% = R225,11$
 OR

Cost = $(6 \times 0) + (9 \times 9,35) + (10 \times 11,16)$
 $= R84,15 + R111,60$
 $= R195,75$
 Including VAT = $R195,75 \times 15\%$
 $= R29,3625$
 $= R195,75 + R29,3625$
 $= R225,11$

Increased block rate tariffs to encourage saving of water 9A
 OR

Also assist small businesses or families with free water 9A
 Accept any other sound reason .

1M
 Cost of first 6 k ■

1M Cost for both 9 and 10
 kilolitres
 1CA Total cost

1M Multiply by 15%
 1CA Cost including VAT

1A
Reason

(6) F
L4

4.2 R0,019 = 1 RWF

$$R? = 745\,614,04 \text{ RWF } 9M$$

$$R? = 0,019 \times 745\,614,04 \text{ 9M}$$
$$= R14\,166,66676 \text{ 9S}$$

$$\text{Bank charges} = 14166,66676 \times \blacksquare\blacksquare$$
$$\blacksquare\blacksquare\blacksquare \text{ 9M}$$

$$= R1\,416,66676 \text{ 9A}$$

$$\text{Andile received} = R14\,166,66676 - 1\,416,66676 \text{ 9 M}$$
$$= R12\,750$$

Statement is valid. 9A

OR

$$\text{Bank charges} = \blacksquare\blacksquare$$

$$\blacksquare\blacksquare\blacksquare \times 745\,614,04 \text{ RWF } 9M$$
$$= 74\,561,404 \text{ 9A}$$

$$\text{Andile received in RWF} = 745\,614,04 - 74\,561,404 \text{ 9M}$$
$$= 671\,052,636 \text{ 9S}$$

$$\text{In Rands: } R0,019 = 1 \text{ RWF}$$

$$R? = 671\,052,636 \text{ RWF } 9M$$

$$\text{Andile received} = R0,019 \times 671\,052,636 \text{ 9M}$$
$$= R12\,750$$

Statement is valid 9CA 1M Concept of ratio
1M

Multiplication

1S Simplification value in R

1M Multiplication of 10%

1A Value of 10%

1M Subtraction

1A
Valid

OR

1M Multiplication of 10%

1A Value of 10%

1M Subtraction

1S Simplification value

1M

Concept of ratio

1M Multiplication

1CA Valid (6) F
L4

9CA 9M

9M 9M 9M 9M

9M

9CA 9CA

8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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4.3.1 Total absentees = 67 9 M
 Absentees on Wednesday =16 9A
 $P(\text{absent on Wed}) = \frac{16}{67}$
 1M Addition to 67
 1A Absentees on Wed
 1CA Fraction
 (3) P
 L2

4.3.2

1A for Monday

boys at 5
 1A for Tuesday girls at 6
 1A for Thurs for girls at 7
 1A for Thurs for boys at 9
 1A for Fri for boys at 7

(5) D
 L2

4.4.1 Value of C:
 $64,2 = \frac{C}{30}$
 $C = 64,2 \times 30$
 $C = 1926$

$C = 1926 - 1853$
 $C = 73$
 Answer invalid
 1M Addition (1853) and
 division by 30

1M Subtraction
 1CA Value of C

1A Invalid (4) D
 L4

4.4.2 $D = 0$ 9A
 No learners scored 30 – 39 marks 9A 1A Value of D
 1A Explanation
 (CA value of D from 4. 4.1
 included in the data) (2) D
 L4

[26]

TOTAL: 100
 024681012
 Monday Tuesday Wednesday Thursday Friday Number of absent learners
 Days of the week Number of absent learners during the week
 Girls Boys 9A
 9A
 Tuesday
 D9A

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P1
EXEMPLAR

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages, including an answer sheet.

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 3

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1.2.3 Explain the meaning of the term ' median '. (2)

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1.3.2 Calculate the percentage loss made on the sale. (2)

[23]

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 2

2.1 Mrs Rogue invested money monthly from 2016 to 2020 at Company A. She terminated her investment and received the statement below from Old Mutual:

Contract number Contracting
party Name of
Investment plan Start date
of contract Last
premium
due date

17801249

R. Rogue Smart MAX

Focused

Education Plan/1

01/07/2016

30/06/2031

Fund value as at 23 March 2020

Investment fund Unit price (cents) Number of
units Fund value

Allan Gray Balanced Fund C Class 8 266,470 A R8 038,07

Total R8 038,07

Difference between fund value and termination value

Fund value R8 038,07
 Reduction fees and transaction charges B
 Termination value R6 995,25
 Withdrawals up to 23 March 2020 R765,57

Expected premium details as at 23 March 2020

Current
 premium
 Annual
 Increase
 rate Next
 premium
 increase New premium Investment
 fund(s) Premium
 Split
 R332,75
 monthly ... 01/07/2020 R366,02 Allan Gray
 Balanced Fund
 C Class 100%
 Total contributions since contract start date
 Total premiums paid R12 924,75

2.1.1 Write down the name of the investment plan that Mrs Rogue has. (2)

2.1.2 Calculate the value of A, the number of units earned from the given fund value.
 (4)

2.1.3 Show by means of calculations that Mrs Rogue lost 45,88% of the total premiums she contributed towards the fund by terminating her investment before the last premium. (4)

2.1.4 Calculate the value (in Rands) of B, the reduction fees and transaction charges.
 (2)

2.1.5 Write down the withdrawal value as at 23 March 2020. (2)

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 5

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2.1.6 Calculate the percentage increase on the monthly premium. Give your final answer to the nearest percentage.

You may use the following formula:

Percentage increase = $\frac{\text{New Premium} - \text{Old Premium}}{\text{Old Premium}} \times 100\%$ (4)

2.2 Alice sells food from a kiosk that she rents in a township. The expenses associated with ingredients and labour for a plate amount to R30.

The formula for expense is: $\text{Expenses} = R500 + R30 \times \text{Number of plates}$

She sells food per plate. The graph below shows her expenses and income for a month.

Use the above information and the graph to answer the questions that follow.

2.2.1 Write down the independent variable in the above context. (2)

2.2.2 Determine the amount Alice pays as fixed expenses. (2)

2.2.3 Write down the formula for the income if a plate is sold at R50, in the form of:
 Income = ... (2)

2.2.4 Determine the profit at the break-even point. (2)

2.2.5 Determine the loss when 8 plates are sold. (4)

[30] 020040060080010001200140016001800

0 5 10 15 20 25 30 35 Amount in Rands(R)

Number of plates ALICE'S EXPENSES AND INCOME FROM SALES

Expenses

Income

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

Copyright reserved Please turn over

QUESTION 3

The Global Wellness Institute did research about the amount of spa revenue generated in different regions of the world in 2017 .

TABLE 1: NUMBER OF SPAS, REGIONS AND THE REVENUE GENERATED

Regions Names of Regions Number of Spas Revenue in Billion dollars (\$b)

P North America 30 394 22,9

Q Latin America-Caribbean 13 856 6,6

R Sub-Saharan Africa 3 984 ...

S Middle East – North Africa 6 057 2,8

T Europe 46 282 33,3

V Asia – Pacific 48 679 26,5

TOTAL

[Source: globalwellnessinstitute.org]

Use the information above to answer the questions that follow.

3.1 Calculate the total number of spas used for the research in 2017. (2)

3.2 Calculate the mean number of spas used. Give your final answer to the nearest whole number. (2)

3.3 Express the number of spas in Europe as a percentage of the total number of spas. (2)

3.4 Determine the number of regions that lies above the range number of spas. (2)

3.5 Write down the unit ratio of spas in regions P and V. (3)

3.6 Calculate the total revenue in billion dollars for the spas if sub-Saharan Africa's revenue is \$5,0 billion less than Latin America-Caribbean's revenue . (4)

3.7 Calculate the median revenue of the regions. (3)

3.8 Determine the probability (as a percentage) of randomly selecting a region with more than 40 000 spas. (3)

[21]

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 7

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QUESTION 4

4.1 GOOD BOYS CAR WASHING BAY is in an area where water is charged according to the water tariffs structure shown in TABLE 2 below.

TABLE 2: WATER TARIFF STRUCTURE

Block Usage in
kilolitre (k ■) Normal Charge per kilolitre
(k■) (Excluding VAT)
1 0 – 6 R0,00
2 +6 – 15 R9,35
3 +15 – 30 R11,16
4 +30 – 45 R12,53
5 +45 – 60 R13,98
6 60+ R15,34

NOTE: VAT is Value Added Tax. The VAT rate is 15%.

Use TABLE 2 above to answer the question that follow.

Calculate how much GOOD BOYS CAR WASHING BAY pays a month including VAT when they use 25 k ■ of water and give a reason why a step up (increasing block rate) system of water tariffs is used to charge water consumption other than a flat single rate. (6)

4.2 Andile received a donation from his brother who works in Rwanda. The donation was 745 614,04 Rwanda Francs (RWF). The bank deducted 10% for bank charges. Andile states that he will receive R12 750. Verify the statement with the necessary calculations .

Use the exchange rate R 0,019 = 1 RWF (6)

4.3 Ibanda High School recorded absentees in the Grade 11 Mathematical Literacy class during the first week when schools reopened after the COVID-19 pandemic break. The data is shown in the table below.

	Monday	Tuesday	Wednesday	Thursday	Friday
Girls	4	6	8	7	10
Boys	5	3	8	9	7

Use the above information to answer the questions that follow.

4.3.1 Determine the probability that a learner chosen at random was absent on Wednesday. (3)

4.3.2 Complete the double bar graph on the ANSWER SHEET provided by plotting the missing bars for the number of boys and girls absent. (5)

8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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4.4 A Grade 11 Mathematical Literacy class at Ibanda High wrote an examination marked out of 100 marks. The results arranged in ascending order are shown below.

23 41 42 50 50 51 54 55 56 57
60 61 65 66 66 67 68 69 70 70
70 72 C 74 76 79 82 85 86 88

Use the above information to answer the questions that follow.

4.4.1 The mean of the above data is equal to 64,2. A learner calculated that the value of C in the above data is 74. Verify , with the necessary calculations, whether the answer is valid. (4)

4.4.2 The frequency table of the above data is shown below.

Class interval	Frequency
20–29	1
30–39	D

40–49 2
50–59 7
60–69 8
70–80 8
80–89 4

Determine the value of D and give a reason for your answer. (2)
[26]

TOTAL: 100

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P1 9
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ANSWER SHEET

QUESTION 4.3.2

NAME OF LEARNER:

GRADE 11:

024681012
Monday Tuesday Wednesday Thursday Friday Number of absent learners
Days of the week Number of absent learners during the week
Girls Boys

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2020

MATHEMATICAL LITERACY P1
EXEMPLAR

MARKS: 100

TIME: 2 hours

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(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 3
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4 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2020)
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17801249
R. Rogue Smart MAX
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01/07/2016
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(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 5

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(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 7
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(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 9

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ANSWER SHEET

QUESTION 4.3.2

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NATIONAL

SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P1
EXEMPLAR

MARKS: 100

TIME: 2 hours

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4 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2020)

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R. Rogue Smart MAX

Focused

Education Plan/1

01/07/2016

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Allan Gray Balanced Fund C Class 8 266,470 A R8 038,07

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premium

Annual

Increase

rate Next

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R332,75

monthly ... 01/07/2020 R366,02 Allan Gray

Balanced Fund

C Class 100%

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Total premiums paid R12 924,75

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(4)

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(2)

2.1.5 Write down the withdrawal value as at 23 March 2020. (2)

(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 5

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2.2.4 Determine the profit at the break -even point. (2)

2.2.5 Determine the loss when 8 plates are sold. (4)
 [30] 020040060080010001200140016001800
 0 5 10 15 20 25 30 35 Amount in Rands(R)
 Number of plates ALICE'S EXPENSES AND INCOME FROM SALES
 Expenses
 Income
 6 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2020)
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 QUESTION 3

The Global Wellness Institute did research about the amount of spa revenue generated in different regions of the world in 2017 .

TABLE 1: NUMBER OF SPAS, REGIONS AND THE REVENUE GENERATED

Regions	Names of Regions	Number of Spas	Revenue in Billion dollars (\$b)
P	North America	30 394	22,9
Q	Latin America -Caribbean	13 856	6,6
R	Sub-Saharan Africa	3 984	...
S	Middle East – North Africa	6 057	2,8
T	Europe	46 282	33,3
V	Asia – Pacific	48 679	26,5
	TOTAL	...	...

[Source: globalwellnessinstitute.org]

Use the information above to answer the questions that follow.

3.1 Calculate the total number of spas used for the research in 2017. (2)

3.2 Calculate the mean number of spas used. Give your final answer to the nearest whole number. (2)

3.3 Express the number of spas in Europe as a percentage of the total number of spas. (2)

3.4 Determine the number of regions that lies above the range number of spas. (2)

3.5 Write down the unit ratio of spas in regions P and V. (3)

3.6 Calculate the total revenue in billion dollars for the spas if sub-Saharan Africa's revenue is \$5,0 billion less than Latin America -Caribbean's revenue. (4)

3.7 Calculate the median revenue of the regions. (3)

3.8 Determine the probability (as a percentage) of randomly selecting a region with more than 40 000 spas. (3)
 [21]

(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 7
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 QUESTION 4

4.1 GOOD BOYS CAR WASHING BAY is in an area where water is charged according to the water tariffs structure shown in TABLE 2 below.

TABLE 2: WATER TARIFF STRUCTURE
 Block Usage in

kilolitre (k■) Normal Charge per kilolitre (k■) (Excluding VAT)
1 0 – 6 R0,00
2 +6 – 15 R9,35
3 +15 – 30 R11,16
4 +30 – 45 R12,53
5 +45 – 60 R13,98
6 60+ R15,34

NOTE: VAT is Value Added Tax. The VAT rate is 15%.

Use TABLE 2 above to answer the question that follow.

Calculate how much GOOD BOYS CAR WASHING BAY pays a month including VAT when they use 25 k ■ of water and give a reason why a step up (increasing block rate) system of water tariffs is used to charge water consumption other than a flat single rate . (6)

4.2 Andile received a donation from his brother who works in Rwanda. The donation was 745 614,04 Rwanda Francs (RWF). The bank deducted 10% for bank charges. Andile states that he will receive R12 750. Verify the statement with the necessary calculations .

Use the exchange rate R 0,019 = 1 RWF (6)

4.3 Ibanda High School recorded absentees in the Grade 11 Mathematical Literacy class during the first week when schools reopened after the COVID -19 pandemic break. The data is shown in the table below.

	Monday	Tuesday	Wednesday	Thursday	Friday
Girls	4	6	8	7	10
Boys	5	3	8	9	7

Use the above information to answer the questions that follow .

4.3.1 Determine the probability that a learner chosen at random was absent on Wednesday . (3)

4.3.2 Complete the double bar graph on the ANSWER SHEET provided by plotting the missing bars for the number of boys and girls absent . (5)

8 MATHEMATICAL LITERACY P1 (EC/ NOVEMBER 2020)

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4.4 A Grade 11 Mathematical Literacy class at Ibanda High wrote an examination marked out of 100 marks. The results arranged in ascending order are shown below.

23 41 42 50 50 51 54 55 56 57
60 61 65 66 66 67 68 69 70 70
70 72 C 74 76 79 82 85 86 88

Use the above information to answer the questions that follow.

4.4.1 The mean of the above data is equal to 64,2. A learner calculated that the value of C in the above data is 74. Verify , with the necessary calculations, whether the answer is valid . (4)

4.4.2 The frequency table of the above data is shown below.

Class interval	Frequency
20–29	1
30–39	D

40–49 2
50–59 7
60–69 8
70–80 8
80–89 4

Determine the value of D and give a reason for your answer . (2)
[26]

TOTAL : 100

(EC/ NOVEMBER 2020) MATHEMATICAL LITERACY P1 9
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ANSWER SHEET

QUESTION 4.3.2

NAME OF LEARNER :

GRADE 11:

024681012
Monday Tuesday Wednesday Thursday Friday Number of absent learners
Days of the week Number of absent learners during the week
Girls Boys

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2017

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 13 pages including an answer sheet.

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2017)
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INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Use the ANSWER SHEET provided to answer QUESTION 5.2 .2.
3. Write your NAME and GRADE on your ANSWER SHEET and hand it in with your ANSWER BOOK .
4. Number the answers correctly according to the numbering system used in this question paper.
5. Start EACH question on a NEW page.
6. You may use an approved calculator (non-programmable and non-graphical), unless

stated otherwise.

7. Show ALL calculations clearly.

8. Round off ALL the final answers appropriately according to the given context, unless stated otherwise.

9. Indicate units of measurement, where applicable.

10. Maps and diagrams are NOT necessarily drawn to scale, unless stated otherwise.

11. Write neatly and legibly.

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P1 3
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QUESTION 1

1.1 Study the payslip below and answer the questions that follow .

Payslip for the period 1 January 2017 – 31 January 2017

Name of
Employer : Date of Payment :

LND Pty Ltd 5 Feb 2017

Name of
Employee : Method of Payment :

Thando Peter Cash

ITEM AMOUNT
(R) OVERTIME
DETAILS

Basic salary 25 820 Overtime hours 5
Total allowances
(breakdown below) 6 455 Total worked
overtime pay R1 290

(i) Transport 3 873 Item Amount
(ii) Uniform 2 582 Other additional
Payments
(Breakdown
below) R38 730

Deductions Annual Bonus R38 730

(i) UIF 258,20
(ii) Income Tax 4 956,38 Net Pay -----
(iii) Loan 2 582
(iv) Pension Fund 1 956,50

1.1.1 Write down the month in which the salary was paid . (2)

1.1.2 Calculate the total deductions for this employee. (2)

1.1.3 Write down the acronym UIF in full. (2)

1.1.4 The employee decided to share R1 290 between his two children Linda and Ludwe in the ratio 3 : 2 respectively. Calculate the amount Linda will receive. (2)

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2017)
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1.2 Asisipho drew the diagram below showing her mother's room.

Calculate the perimeter of the room in centimetres . (2)

1.3 Study the strip map below and answer the questions that follow .

1 : 250 000

1.3.1 Write down the compass direction of Lauterbach from Hanau. (2)

1.3.2 Explain the meaning of 1 : 250 000 on the chart . (2)

1.4 Bonny recorded Graaff Reinet 's temperatures in degrees Celsius from Wednesday, 13th of September to Friday, 22nd of September 2017. Study the information below and answer the questions that follow .

Dates 13 14 15 16 17 18 19 20 21 22

Min. 8 8 11 7 7 9 12 12 12 12

Max. 25 21 26 17 19 23 29 28 26 27

[Source: accuweather.com]

1.4.1 Arrange the maximum (max .) temperatures from smallest to biggest. (2)

1.4.2 Write down the lowest minimum value of the data. (2)

1.4.3 Write down the meaning of the " mode " in a data. (2)
[20]

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QUESTION 2

2.1 Quzini High School soccer and rugby teams are planning a tour to Port Elizabeth. The team's manager, Lwando obtained a quotation for accommodation costs per night at Red Lodge shown in TABLE 1 below .

TABLE 1: RED LODGE ACCOMMODATION COSTS

Type of room Cost per night per unit

2 Sleeper R855

4 Sleeper R1 035

6 Sleeper R1 400

8 Sleeper R1 495

Executive (Honey moon) R1 450

NOTE : These prices exclude VAT (VALUE ADDED TAX = 14%)

Use the table above to answer the questions that follow .

2.1.1 Write down the cost for the executive room. (2)

2.1.2 Express the cost for the executive room as a percentage of the cost of a room where 8 sleep. Give your answer to ONE decimal point. (4)

2.1.3 Calculate the total amount to be paid per day for accommodation if the learners use the rooms as follows:

1 room where 8 sleep

2 rooms where 4 sleep

3 rooms where 6 sleep (4)

2.1.4 Hence, calculate the total cost including Value Added Tax (VAT of 14%) of accommodation if the learners will stay for four days and return on the fifth day. (5)

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2017)

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2.2 A deposit of 12,5 % of the total accommodation cost excluding VAT is required to secure the booking and the balance is payable on the day of arrival.

2.2.1 Calculate the amount that must be paid as deposit. (3)

2.2.2 Calculate the balance amount to be paid on the day of arrival. (2)

2.2.3 Write in words the balance amount to be paid on the arrival day. (2)

The school received a donation of 20 US Dollars (\$) from an ex-rugby player to cater for transport.

2.2.4 Convert the \$ 20 to rands.

Given \$ 1 = R14, 2058. (3)

2.3 Study the parking tariffs below and answer the questions that follow.

MALL A: PARKING TARIFFS

Monday – Thursday Friday, Saturday, Sunday and Public Holidays

0 hour – 1,5 hours	R0,00	0 hour – 1,5 hours	R5,00
1,5 hours – 2,5 hours	R9,00	1,5 hours – 2,5 hours	R10,00
2,5 hours – 3,5 hours	R12,00	2,5 hours – 3,5 hours	R10,00
3,5 hours – 4,5 hours	R14,00	3,5 hours – 4,5 hours	R10,00
4,5 hours – 5,5 hours	R16,00	4,5 hours – 5,5 hours	R10,00
5,5 hours – 6,5 hours	R19,00	5,5 hours – 6,5 hours	R10,00
6,5 hours +	R35,00	6,5 hours +	R30,00
Lost ticket	R35,00	Lost ticket	R35,00

Quzini High School's bus is expected to park at Mall A's parking area during the week. The parking is for 2,6 hours daily, from the 2nd to the 5th day of the tour while the learners enjoy a breakfast. Calculate the total amount of money to be paid for parking. (2)

[27]

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P1 7

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QUESTION 3

3.1 Dogs are fed according to their weight to keep them healthy. Nikki keeps two dogs

, Tiger and Tommy, which weigh 22 kg and 42 kg respectively.

Study the dog food specifications label in TABLE 2 below to answer the questions that follow.

TABLE 2: DAILY FOOD REQUIREMENTS ACCORDING TO WEIGHT

Size of dog	Weight of dog	Daily food requirement
Small	Up to 20 kg	250 g
Medium	20 – 40 kg	Minimum 250 g – Maximum 500 g
Large	40 kg+	Minimum 500 g – Maximum 750 g

3.1.1 Calculate the amount of daily minimum food requirements for the two dogs. Give your answer in cups.

Use 1 cup = 125 g (3)

3.1.2 Calculate the number of days a 10 kg packet of dog food will last the two dogs feeding them with minimum requirement ratios .

You may use the formula :

Number of days = weight of a packet
daily requirement of two dogs (4)

3.1.3 Convert Tommy's weight to tons. 1 ton = 1 000 kg. (2)

8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2017)

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3.2 Thandi wants to lay ceramic tiles in her rectangular bedroom that measures 3 m by 3,5 m. Study the details of the size and the packaging of the tiles below and answer the questions that follow .

Dimensions of a tile Cost of tiles

[Adapted from BTM catalogue]

3.2.1 Convert 43 cm to metres (m) (2)

3.2.2 Calculate the area of Thandi's main bedroom.

You may use the formula:

Area of a rectangle = length x breadth (2)

3.2.3 Calculate the number of boxes of tiles needed for Thandi's bedroom. (tiles are sold in boxes only)

You may use the formula:

Number of boxes of tiles = Area of the bedroom
area of a tile x number of tiles in a box (5)
[18]

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QUESTION 4

4.1 Study the floor plan of the shopping mall below and answer the questions that follow .

[Source: www.decorfarms.me]

4.1.1 Determine the number of 2 x 3 units. (2)

4.1.2 Identify the size of units found at the corners of the mall. (2)

4.1.3 Lindi entered the mall on the West side. She went straight through Region V and turned left before entrance 3, passing three shops on the right and entered the fourth shop on the left . Write down Lindi's destination. (3)

10 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2017)

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4.2 Study the second last stage of assembling a learner's desk below and answer the

questions that follow .

SECOND LAST STAGE OF ASSEMBLING THE SCHOOL DESK

Determine the following:

4.2.1 The number of screws needed to fix two of these school desks (2)

4.2.2 The direction in which the screw is tightened . Is it clockwise or anticlockwise? (2)

[11]

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P1 11

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QUESTION 5

5.1 TABLE 3 below shows the information about the dams that had above ten million cubic metres of water in January 2016. The terms “fullest and emptiest ” are used to indicate the amount of water in the dam in relation to its capacity.

TABLE 3: DAMS WITH MORE THAN TEN MILLION CUBIC METRES OF WATER

Province Fullest dam

name Fullest

dam

capacity

(in million

cubic

metres) Emptiest dam

name Emptiest dam

capacity

(in million

cubic metres)

Northern Cape Boegoeberg 19,9 Spitskop 57,9

Western Cape Garden

Route 10 Gamkapoort 36,3

Free State Saulspoort 15,7 Krugersdrift 71,5

North West Bospoort 15,8 Setumo 20,8

Gauteng Roodeplaat 41,2 Bronkhorstspuit 57

Limpopo De Hoop 348,7 Glen Alpine 18,9

Mpumalanga Heyshope 445 Ohrigstad 13,5

KwaZulu

Natal Inanda 237,5 Klipfontein 18,1

Eastern Cape Mthatha 224,7 Darlington 180,9

Use TABLE 3 to answer the questions that follow .

5.1.1 Identify the fullest dam with the maximum capacity . (2)

5.1.2 Calculate the difference between the highest fullest dam and the highest emptiest dam. (3)

5.1.3 Calculate the mean emptiest dam capacity. (3)

5.1.4 Write down the name(s) of the emptiest dams with less than twenty million cubic metres. (2)

5.1.5 Determine the range capacity for the fullest dams. (2)

5.2 Study TABLE 4 that shows the number of absent Grade 11 learners at Siyo High School and answer the questions that follow .

TABLE 4: Number of Grade 11 absentees for two weeks:

Week 1														
dates	4th	5th	6th	7th	8th									
No. of absent learners	22	1	2	4	18									
Week 2														
dates	11th	12th	13th	14th	15th									
Number of absent learners	8	12	3	5	14									

5.2.1 Identify the week which had the highest number of absentees. (2)

5.2.2 Use a bar graph to represent the number of learners who were absent in Week 2 . (5)

5.3 Bathandwa had a bag with eighteen balls. Three balls were white, five green and the rest were blue.

5.3.1 Determine the probability that the ball randomly selected from the bag was green. (2)

5.3.2 Express the probability of randomly selecting a blue ball from the bag as a percentage. Your answer should be rounded off to two decimal places. (3)

[24]

TOTAL : 100

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P1 13

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ANSWER SHEET

NAME OF LEARNER:

GRADE 11: _____

QUESTION 5.2

0246810121416

11th 12th 13th 14th 15th Number of absent learners

2nd week of September 2017 Grade 11 absent learners during the 2nd week of September 2017

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2016

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 11 pages, including 1 answer sheet and 1 annexure .

Imlit1

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2016)

Copyright reserved Please turn over INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.

Use the ANNEXURE to answer QUESTION 1.

Answer QUESTION 4. 7 on the ANSWER SHEET provided and hand it in with your ANSWER BOOK.

2. Number the questions correctly according to the numbering system used in this question paper.

3. A non-programmable and non-graphical calculator may be used, unless stated otherwise.

4. ALL calculations and steps must be shown clearly.

5. Units of measurement must be indicated where applicable.

6. Start each question on a NEW page.

7. Round all final answers appropriately according to the context used.

8. Write neatly and legibly.

(EC/NOVEMBER 2016) MATHEMATICAL LITERACY P1 3

Copyright reserved Please turn over QUESTION 1

1.1 Sandra paid cash for the service of her car at Bennie's Service station. Study the information in the ANNEXURE to answer the questions that follow:

1.1.1 Name the Franchise Dealer that serviced Sandra's car. (2)

1.1.2 Calculate the value of A, the amount charged for the labour. (2)

1.1.3 Show, with calculations, that the discount for the battery is 50%. (2)

1.1.4 Calculate the total unit price of all the parts used for servicing the car. (2)

1.1.5 Calculate the value of B, the amount of VAT paid by Sandra. (2)

1.1.6 Calculate the difference between the highest unit price and lowest unit price. (3)

1.2 Study the extract from Sandra's cheque account statement below and answer the questions that follow: (Some information is omitted.)

Table 1: Sandra's cheque account statement from 23 to 25 March 2016

Date	Description	Amount
(R)	Balance	(R) Bank
	Accrued	
	Charges	

23 March Opening Balance R22 598,09
 23 March Cell transfer to card 2 000,00 24 598,09
 23 March Airtime Top -up - Airtime
 0633148090 110,00 24 486,99 1,10
 24 March POS Purchase Cheque
 Card @ CFR Shop 539,32 23 947,67
 24 March ATM Cash - Main street 4 500,00 19 447,67
 24 March Airtime Top -up - Airtime
 0732729000 55,00 19 391,57 1,10
 24 March Send money to
 0633148090 500,00 18 882,07 9,50
 25 March Cheque Card Fuel
 Purchase @ Ted Garage 300,00 18 582,07
 Closing balance 18 582,07

1.2.1 Explain the term “ opening balance ”. (2)

1.2.2 On the 24th of March 2016 the bank accrued charges for the amount sent to 0633148090 was R9,50. Determine the bank accrued charges as a percentage of the amount sent. (3)

1.2.3 Determine the total amount Sandra paid for using the bank to buy airtime. (3)

1.2.4 Write down Sandra's balance on the 25 March 2016 in words. (2)
[23]

4 MATHEMATICAL LITERACY P1 (EC/NOVE MBER 201 6)

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2.1 Nonine is a retired teacher who usually plants vegetables in her rectangular garden. She sent her soil sample to the laboratory to be tested. The results from the sample indicated the need to apply 2 pounds of 15 -15-15 (Nitrogen - Phosphorus -Potassium) fertiliser per 100 square feet (■■■2). She plans to spread the fertiliser over a 20 x 15 foot area for corn. She has three 10 -foot rows ; one each for carrots, lettuce and spinach. She also has 10 tomato plants, which are fertili sed individually.

Use the picture below that shows Nonine's rectangular garden and the conversion table given to answer QUESTION 2.1

10 foot 10 foot
 10 foot

1 ■■=3,281 ■■
 1 ■■■■■■ =2 ■■■■■
 1 ■■■■ =250 ■■
 1■■2 =10,764 ■■■2

2.1.1 Write down the length and the width of Nonine's garden in feet. (2)

2.1.2 Calculate the area of Nonine's vegetable garden. Give your answer in square metres.

You may use the formula: Area of a rectangle = length x breadth (4)

25 ft 30 ft
 (EC/NOVEMBER 2016) MATHEMATICAL LITERACY P1 5

Copyright reserved Please turn over 2.1.3 Calculate the amount of the fertiliser needed for the corn area in the garden. Give your answer in pounds.

You may use the formula:

$$\frac{\text{Number of items}}{\text{Total number of items}} = \frac{\text{Number of items}}{\text{Total number of items}} \quad (5)$$

2.1.4 Each tomato plant needs 0,15 pound of the fertiliser. Convert the amount of the fertiliser needed for each tomato plant into cups. (3)

2.2 Nonine's daughter, Mimmy, fell sick and consulted a doctor. She was given medication and instructed to complete an antibiotic treatment. The table below shows the medication given to her.

Table 2: Certified copy of doctor's script (some information is omitted)

Code	Type of medication	Volume	Amount
797979018	Flusin DM	Take two medicine measures (10 m ■)	
	Three times a day	Syrup 100 m ■	
10D R67,46			
756156017	Ponstan 50 mg/5 m ■		
	Take three medicine measures (15 m■)	four times a day 200 m ■	10D R37,70
710041002	Augmentin ES 600	Take two medicine measures (10 m ■) twice a day**complete course**	100 m ■ 5D 710041002

2.2.1 Calculate the total number of millilitres of medicine measures that Mimmy had to take in the morning at breakfast and in the evening.at supper. (3)

2.2.2 Calculate the amount of Augmentin ES 600 to be taken on the last day. (2)

[19]

6 MATHEMATICAL LITERACY P1 (EC/NOVE MBER 201 6)

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3.1 Study the map below of Fraser River Area from Haney to the Mouth and answer the questions that follow:

[Source: www.env.gov.bc.ca]

3.1.1 Determine the number of sediment monitoring stations from the map. (2)

3.1.2 Write down the type of monitoring station named Marion (Jacobs) Lake. (2)

3.1.3 Identify the sediment monitoring station that lies between Main Arm and North Arm to join Fraser. (2)

3.1.4 Measure the distance (in a straight line) in centimetres between Haney and NA -2. (2)

3.1.5 Use your measured distance in QUESTION 3.1.4 and the given scale to determine the actual distance between Haney and NA -2 in kilometres. (3)

Copyright reserved Please turn over 3.2 The Vellem family invited few guests to celebrate in their son's graduation ceremony for completing a Diploma in Management from Walter Sisulu University. The hall was partitioned into two by a curtain making table D and G unavailable to guests.

Study the hall plan below and answer the questions that follow:

Tables

A-I, Head Table and Gift table

-seat occupied by guests and family members

3.2.1 One of the guests did not want to sit close to the window. Identify the table where this guest may be seated. (2)

3.2.2 Give the general compass direction of Table F from Table H. (2)

3.2.3 Determine the probability that a person randomly chosen will be seated at tables D and G during the ceremony. (2)

3.2.4 One of the guests was asked to walk straight from the door to sit at the second left table. Name the table where this guest will be seated. (2)
[19]

N



8 MATHEMATICAL LITERACY P1 (EC/NOVE MBER 201 6)

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Table 3 below compares prices of selected food items in the rural and urban areas for January 2015. Study the table and answer the questions that follow:

Table 3: Comparison between urban and rural food prices (selected food items)

Product Rural food prices

January 2015

(in Rands) Urban food prices

January 2015

(in Rands) Price difference

Rand per unit

Full cream Long Life Milk (1 ■) 12,03 12,59 0,56

Loaf of Brown Bread (700 g) 9,57 10,29 0,72

Loaf of White Bread (700 g) 10,31 11,42 1,11

Maize Meal (5 kg) 32,49 33,73 1,24

Margarine(500 g) 17,89 21,68 3,79

Rice (2 kg) 23,62 23,45 -0,17

Sunflower Oil (750 m■) 14,59 17,25 2,66

Ceylon/Black Tea (62,5g) 9,89 9,68 -0,21

White Sugar (2,5 kg) 29,63 26,31 -3,32

Average A

[Source: Stats SA, 2015]

4.1 Arrange the rural food prices in descending order. (2)

4.2 Determine the median for rural food price s. (2)

- 4.3 Calculate the range for urban food prices. (3)
- 4.4 Identify the minimum price difference in rand per unit. (2)
- 4.5 Identify ONE product with the largest price difference between Urban and Rural area. (2)
- 4.6 Identify the mode for urban food prices. (2)
- 4.7 Draw a bar graph to show the last FIVE products from urban food prices. Use the graph paper provided on ANSWER SHEET 1 . (6)
- 4.8 Write down your observation about the food prices for both urban and rural customers. (2)
- 4.9 Calculate the value of A, the average price difference in rand per unit. (3)
- 4.10 Determine the probability that a product randomly selected has a mass of 700 g. Give your answer to the nearest percentage. (3)

[27]

(EC/NOVEMBER 2016) MATHEMATICAL LITERACY P1 9

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Victoria went to Sparrow shop to buy few items. She has a loyal card that she uses to earn points when the card is presented at the till. The earned points can be converted to cash to buy some items from the shop when it is needed . Study the till slip below she received from Sparrow and answer the questions that follow: (*some information is omitted*)

SPARROW SHOP
 Roseplace.Tel: 716 1610
 Customer: Victoria Prices (R)
 Parm Stew F/Cus 1 kg 22,49
 Mc Beans (Fresh Greenbeans) 1 kg 36,99#
 A/Chix B/P 4 -PCE 0,542 kg @ 53,99 p/kg 29,26
 A/Chix B/P 4 -PCE 0,474 kg @ 53,99 p/kg 25,59
 Flora regular margarine per kg 2 @ 1 kg 99,98
 Sunday Times National 1's 22,00
 Bostik Prestik L 100 g 17,99
 Koo PCH.Slice SR 410 g 9,99
 Valpre Still 1,5 ■ 13,49
 Carrier bag 2 @ 0,50 24 ■ 1,00
 TOTAL FOR 12 ITEMS 278,78
 Tendered debit card 478,78
 Change cash 200,00
 Points earned:
 Before This sale: 1639
 This sale: 13
 Total points : 1652
 VAT rate Excluding. Tax Included
 0,00% 36,99 0,00 36,99*
 14,00% 212,10 29,69 241,79
 Slip Till Cashier Date Time
 4372 005 21 03.04.16 10:00

- 5.1 Identify the price of an item that is zero VAT (Value Added Tax) rated. (2)
- 5.2 Identify the cash amount Victoria asked from the cashier at the till in Sparrow shop. (2)
- 5.3 Calculate the value of one point earned by customers at Sparrow Shop. Give your answer to the nearest ten rand. (2)

5.4 Calculate the cost of 600 g of Bostik Prestik at Sparrow shop excluding VAT. (2)

5.5 Calculate the price of 1 kg Flora Regular margarine. (2)

5.6 Calculate the total for all food items on which VAT is charged. (2)
[12]

10 MATHEMATICAL LITERACY P1 (EC/NOVE MBER 2016)

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QUESTION 1

Bennie's Service Station
Franchise Dealer for Mercedes
Benz, Chrysler, Jeep, Dodge,
Mitsubishi, Freightliner and
FUSO vehicles
Buffalo City
Reg.No. TM 1234/0 VAT Reg.No.10465 25987
Billy Road, P.O. Box 238, East London, 5200
Tel.(043) 745 1254 Fax. (043) 745 2334

Postal Address/ Posadres Cash Tax Invoice Deliver to/Gelewer aan
B/Bay cash sales Invoice Number Mrs. Sandra Twalo
10 Lily Street
Sunset Bay
5201
23215874
Op Vehicle
76416
REGISTRATION TDZ 652 EC Date of
1st reg. 26/04/11 Doc no. 26304/
132568
MODEL Mercedes
Benz Selling
dealer 25A Date 24/03/16

Operation/Part no./Description Quantity/
Time units Unit
Price (R) Unit Amount
Consumables 0100MSH
Engine Oil U 5,50 29 159,50 S
MA271 180 05 09 Filter Oil X 1 212,82 EACH 106,41 S
MA 001 986 19 71/ 1075 Cleaner
Solv X 1 4,99 EACH 2,50 S
MA 000 989 08 07/13 Fluid X 1 75,87 EACH 37,94 S
MA 204 890 2311 TS Wiper Blade X 1 697,13 EACH 348,57 S
MA 201 565 456 03 Battery X 1 2469,92 EACH 1234,96 S
MA 313 836 0319 Filter Combi X 1 726,16 EACH 363,08 S

Goods
value VAT
RATE Taxable amount
(incl labour) NET TOTAL 3390,96
VAT/BTW B
Parts (X) 2093,46 14 % 3390,96 Rounding 0,09
Labour A
Sublet 0,00
Deposit 0,00 TOTAL 3865,60

Goods are supplied subject to ownership vesting in the seller until paid for in full.

Copyright reserved Please turn over ANSWER SHEET

SURNAME AND NAME

GRADE 11:

QUESTION 4.7

The graph showing selected urban food prices

0510152025303540
Maize Meal Margarine Rice Sunflower Ceylon Tea White Sugar Prices (Rands)
Selected food items

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2015

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 16 pages including 1 answer sheet and 2 annexures .

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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INSTRUCTIONS AND INFORMATION

1. This paper consists of FIVE questions. Answer ALL the questions.
QUESTION 3.1.3 must be answered on the ANSWER SHEET .
Use ANNEXURE A to answer QUESTION 2.2 and ANNEXURE B to answer QUESTION 4.

2. Number your answers correctly according to the numbering system used in the question paper.

3. You may use an approved (non-programmable and non-graphical) calculator , unless stated otherwise.

4. ALL calculations and steps must be shown clearly.

5. Units of measurement must be indicated where applicable.

6. Start EACH question on a NEW page.

7. Write neatly and legibly.

8. All answers must be rounded up, down or according to the context used, unless stated otherwise.

1.1 Anita runs a business and she wants to buy the cellphone for her personal use. She visited two cellphone companies to compare the prices of the same phone. The findings are listed in the tables below as follows:

Nokia Lumia 630

Off peak : 20:00 (06:59:59 (Monday to Friday)
Peak time: 07:00 –19:59:59 (Monday to Friday)
Weekends : Friday 20:00 –Monday 06:59:59 and Public holidays

1.1.1 Calculate the total cost of this cellphone from Company A for the duration of contract excluding calls and SMS es if she also took insurance. (4)

1.1.2 Calculate the total cost of a n 8 minute cellphone call from Company A made at 20:25 on a Friday. (2)

1.1.3 Calculate how much will the cellphone following year from Company B cost in cash , if the inflation rate is 4,4% per year. (2)

Company A
R169 PM x 24
R100 anytime PM
Peak time SMS R0,50
Off peak SMS R0,50
Peak time calls R0,79
Off peak calls R0,79
Cash: R1 899
Optional insurance
@R55 per month
Company B
R169 PM x 24
50 anytime minutes
50 SMS es PM
100 MB data P M
Peak time SMS R0,80
Off peak SMS R0,35
Peak time calls R1,30
Off peak calls R1,12
Cash: R1 999
Optional insurance
@ R55 per month

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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1.2 Anita's personal calls were recorded as follows for the first week of March 2015.

Times Peak Off peak SMS s

Duration of a call 18 calls at
2 minutes per
call 15 calls at
3 minutes per
call 25 SMS es off peak
10 SMS es peak
hour

NOTE: All the calls and SMS es have been c orrected to the nearest minute.

1.2.1 Calculate the total amount she will pay for the off peak SMS es for the first week if she uses Company B. (2)

1.2.2 Determine the ratio of the calls made during peak time , to that of the calls made during off peak time. (2)

1.2.3 Determine the probability that the SMSe s made by Anita are at off peak time. Give your answer in its simplest form. (2)

1.3 Anita joined a stokvel group of 8 women and started paying contributions in January 2014 and each member has to contribute R400,00 a month. Each member had to raise an amount of R2 000,00 excluding monthly contributions. They will share the money amongst each other on the 10th of December 2014. The last month for contributions is November 2014. If you borrow any amount the interest charged is 20% of the amount loan debt per month. The loan increases by the interest amount if not paid. The interest raised by a member from a loan will belong to her.

1.3.1 Anita took a loan of R1 525, 00 for two months. She did not pay the interest in the first month. Calculate the total interest amount she will have to pay at the end of two months. (2)

1.3.2 Determine the total amount they will have as a group, including Anita's additional interest amount. (3)

1.3.3 Calculate the total amount Anita will receive on the day they share their money , including her own additional interest amount. (3)

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 5

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1.4 Anita saw a book from Ports SA advertising boat cruise s. Hazel, her daughter got it from the Marine Expo at the Mdantsane Indoor Sports Centre. She chooses the in side cabin price of the cruise from Barcelona to Venice. It is for 9 nights and the cost is from \$1,587 to \$4, 760.

1.4.1 Calculate the maximum cost for cruising in an in side cabin of the ship in rands. (1\$ = R 11,82 .) (2)

1.4.2 There was a helicopter at the expo. It costs R1 571,05 for a 30 minute flight excluding VAT. Calculate the cost of the flight for an hour including VAT. (3)

[27]

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 2

2.1 Mr Mambo won a 68 km marathon from Bisho to East London in 2014 . He was a member of Maxed Elite Running Club. He received a cash prize. In 2013 the winner complete d the race in 3 hours 50 minutes and 36 seconds.

Male Running Club Time

1. Marko Mambo Maxed Elite 04:07:05
2. Elias Mabane Nedbank 04:10:16
3. Peter Muthubi Running 04:12:40

[Source: Eastern Cape Today (10 October –17 October 2014)]

2.1.1 Write down the number that appeared on Mr Mambo's running shirt. (2)

2.1.2 Convert Mr Mambo 's running time to hours. Give your answer to the nearest hour. (3)

2.1.3 Calculate Mr Mambo's average speed in kilometres per hour.

You may use the formula:

Average speed =
(3)

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 7

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2.2 Use ANNEXURE A to answer the following questions .

2.2.1 Write down the 47014 train arrival and departure times at Springfontein. (2)

2.2.2 Identify the station where train 73014 arrives during the first hour of the day. (2)

2.2.3 Study the logo of the Passenger Rail Agency of South Africa as displayed below and answer the questions that follow.

Sketch the logo of the Passenger Rail Agency of South Africa.

Radius of a bigger circle = 5 m

Radius of a smaller circle = 250 cm

Calculate the area of the shaded part in m².

You may use the formula:

Area of a circle = , where = 3,142 (5)
[17]

250 cm

8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 3

3.1 Study the pie charts below and answer the questions that follow.

FIGURE 1: PERCENT DISTRIBUTION OF FIVE LEADING CAUSES OF DEATH, BY AGE GROUP: UNITED STATES, PRELIMINARY 2011

[Source: National Vital Statistics System: Mortality]

3.1.1 Determine the total number of people aged 25 –44 dying from accidents. (2)

3.1.2 Identify the cause s of death amongst 1 –24 year -old people that are less than 10%. (2)

3.1.3 Represent the percentages of causes of deaths for age group 25 – 44 years on a line graph. Use ANSWER SHEET 1 that is provided. (6)

3.1.4 Calculate the difference in the total number of deaths of the two age intervals. (2)

3.1.5 Identify the age interval that has high mortality percentage in heart disease. (2)

3.1.6 Give a reason why the accident percentage for the age group 1–24 is higher than 25 –44 age group. (2)

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3.2 The table below represents the 2014/15 National First Division results of the teams that play in it. Study the table below and answer the questions that follow.

TABLE 1 : NATIONAL FIRST DIVISION TEAMS AND THEIR RESULTS

Teams P W D L GD Points

1. Golden Arrows	29	15	12	2	20	57
2. Jomo Cosmos	29	14	8	7	22	50
3. Black Leopards	29	14	6	9	11	48
4. Thanda Royal Zulu FC	29	13	8	8	14	47
5. Cape Town All Stars	29	12	11	6	8	47
6. Milano United	29	9	15	5	7	42
7. Vasco da Gama	29	9	12	8	3	39
8. FC Cape Town	29	9	12	8	0	39
9. Baroka FC	29	9	10	10	5	37
10. Royal Eagles	29	10	7	12	-12	37
11. Highlands Park	29	10	6	13	0	36
12. African Warriors	29	9	9	11	-9	36
13. Santos	29	10	6	13	-11	36
14. Witbank Spurs	29	7	10	12	-10	31
15. Garankuwa United	29	5	7	17	-18	22
16. Maluti FET College	29	5	5	19	-30	20

P = Played

D = Drawn

GD = Goal Difference

D = Drawn

L = Lost

[Source: www.kickoff.com/mobile]

3.2.1 Calculate the average points won by the listed teams. (3)

3.2.2 Determine the mode total of the points earned by these teams. (2)

3.2.3 Identify the team that has the least points on the table. (2)
[23]

10 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 4

4.1 The learners enjoy the bus ride every morning when going to school provided by the Department of Transport (Scholar Transport). Study the diagram showing the seating arrangement inside the school bus.

FIGURE 2: SEATING PLAN INSIDE A SCHOOL BUS

N

4.1.1 Calculate the number of passenger seats displayed inside the bus. (2)

4.1.2 On the first day the learners were given tickets in order to get to the bus. Nandi came late carrying a ticket B14. Give Nandi directions to get to her seat as she enters the door. (3)

4.1.3 Soso left his books in the fourth row on the western side of the seat (next to the window). Identify the seat number where Soso's books are. (2)

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 11

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4.2 Use the map in ANNEXURE B to answer the questions that follow .

4.2.1 Identify the type of map displayed in ANNEXURE B. (2)

4.2.2 Calculate the distance between Polokwane and Hammanskraal. (2)

4.2.3 One road will let you pass through the most number of toll plaza symbols (T) between Johannesburg and Beitbridge. Name the road and the number of toll gates. (2)

4.2.4 Give the compass direction to Centurion from Devon. (2)
[15]

12 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 5

5.1 Thabo sees an advertisement for a fridge at Bill's Furniture Shop and wants to buy it. He has R5 100,45 in his bank account.

5.1.1 Identify the capacity of the advertised fridge in milliliters. (3)

5.1.2 Write down the meaning of "interest rate". (2)

5.1.3 Calculate the amount Thabo owes the furniture shop after the deposit has been paid. (2)

5.1.4 Calculate the total interest Thabo will pay, if Bill's Furniture Shop charges simple interest at a rate of 21,75% per annum on the balance after the deposit has been paid. (3)

Cash Price: R4 799,00

Hire Purchase:

Deposit: R479,90 and
pay the balance over 24
months.

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 13

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5.2 The following data was taken from 15 Grade 11 Mathematics Learners School Based Assessment (SBA) mark in percentages.

Learners	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
% obtained	32	35	36	37	42	46	49	46	46	58	62	64	66	67	38

5.2.1 Determine the SBA median percentage. (2)

5.2.2 Determine the interquartile range of the SBA percentages. (3)

5.2.3 Determine the probability that a learner randomly selected will have a percentage of more than 40%. Give your answer in a simplified fraction. (3)

[18]

TOTAL: 100

14 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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ANSWER SHEET 1

NAME AND SURNAME :

GRADE 11:

Types of	death
Accidents	cancer
Heart	disease
Suicide	Homicide
All other causes	% death

Percentages of 5 leading causes of death
(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 15
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NAME AND SURNAME:

GRADE 11:

ANNEXURE A

16 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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ANNEXURE B

[Source: www.aa.co.za]

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MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages and 1 annexure.

MATHEMATICAL LITERACY P1

EXEMPLAR 2013 NATIONAL

SENIOR CERTIFICATE

GRADE 11

Mathematical Literacy/P1 2

NSC – Grade 11 Exemplar DBE/2013

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Answer QUESTION 5. 1.4 on the attached ANNEXURE. Write your name and class in the spaces on the ANNEXURE and hand in the ANNEXURE with your ANSWER BOOK.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Start EACH question on a NEW page.
5. You may use an approved calculator (non-programmable and non-graphical) unless stated otherwise.
6. Show ALL the calculations clearly.
7. Round off ALL the final answers to TWO decimal places, unless stated otherwise.
8. Indicate units of measurement, where applicable.

9. Maps and diagrams are NOT necessarily drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

Mathematical Literacy/P1 3

NSC – Grade 11 Exemplar DBE/2013

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QUESTION 1

1.1 1.1.1 Without rounding off, convert 2,35 m to mm. (1)

1.1.2 A 3 kg packet of onions costs R19,99.

Calculate the costs of onions per kg.

(2)

1.2

Spark 's Butchery is a popular butchery in Karkloof. The butchery supplies meat to other butcheries in Howick and Pietermaritzburg.

Spark 's Butchery ask s the following prices for some of the ir products:

Product	Cost price per kg	Selling price per kg
---------	-------------------	----------------------

Lamb chops	R54,00	R85,00
------------	--------	--------

Fillet steak	R76,00	A
--------------	--------	---

Spare ribs	R43,00	R53,75
------------	--------	--------

NOTE: Prices include 14% VAT.

1.2.1 Calculate the percentage mark -up that Spark's Butchery will have on spare ribs .

Use the formula:

Percentage mark -up =

$\frac{\text{Selling price} - \text{Cost price}}{\text{Cost price}} \times 100\%$

(3)

1.2.2 Calculate how many kilograms of lamb chops Spark 's Butchery could buy for R2 000. (2)

1.2.3 Determine the value of A, the selling price of the fillet steak, if it has a mark -up of 30%.

(3)

1.2.4 A customer buys 1,2 kg of spare ribs and 0,5 kg of lamb chops. Calculate the total amount paid by the customer.

(4)

1.2.5 Calculate the profit on the sale of 50 kg of spare ribs. (4)

[19]

Mathematical Literacy/P1 4

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QUESTION 2

Yolanda wants to buy the home theatre system seen in the advertisement below.

She chooses to buy it on hire purchase.

2.1 Calculate the deposit paid in rand. (2)

2.2 Show, through calculation, that the total amount to be paid, excluding the deposit, is

R7 306,19 if the store charges simple interest at a rate of 11,5% per annum on the balance after the deposit has been received.

(4)

2.3 Calculate the amount Yolanda must pay each month (the monthly instalment). (2)

2.4 The home theatre's woofer has a length of 8 inches, a height of 12 inches and a width of 6 inches.

If 1 inch = 2,54 cm, determine the height of the woofer in centimetres.

(2)

2.5 Yolanda budgets an amount of R300 per month for the purchase of CDs and DVDs to play on the system.

2.5.1 Calculate how much Yolanda will spend on the purchase of CDs and DVDs per year.

(2)

2.5.2 She buys CDs at an average of R120 per month and DVDs at an average of R150 per month and uses the rest of her budget to buy other accessories.

Calculate how much she will have left per month to buy other accessories.

(3)

[15]

woofer Home Theatre System

Cash Price: R6 599,99

OR

Hire Purchase:

10% deposit and pay the balance over 24 months.

(Total amount to be paid excluding the deposit is

R7 306,19)

Mathematical Literacy/P1 5

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QUESTION 3

3.1 Ms Senokoane is a squash coach at Katlego High School. She needs to repaint the walls of

the squash court as well as renew the lines marking the various regions of the court.

Below is a 3-D scale drawing of a squash court.

Use the information in the diagram above and answer the questions that follow.

3.1.1 Write down the height of the tin section in cm. (2)

3.1.2 Calculate the distance, in mm, from the service line to the out line against the front wall. (2)

3.1.3 Calculate the area, in m², of the service box.

Use the formula: Area = length × breadth
(3)

4830

Right and back walls removed for clarity. All measurements given in millimetres. All lines are 50 mm thick.

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3.1.4 The front wall and the two identical side walls are to be painted white, with the exception of the tin section, which is to be painted black.

Calculate the area, in m², that must be painted white.

Use the following formula:

Area to be painted = $k(2p + b) - b \times t$, where

k = height of the squash court, including out line section

p = length of the squash court

b = breadth of the squash court

t = height of the tin section

(6)

3.2 The type of paint to be used is such that 1 litre of paint covers 8 m².

The white paint is sold in 5 litre containers.

3.2.1 Determine, to the nearest litre, the amount of white paint required to paint the walls, excluding the tin section.

(3)

3.2.2 Calculate the cost of the white paint if one 5 litre container of paint costs R215,85.

(4)

[20]

Mathematical Literacy/P1 7

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QUESTION 4

Clyde lives in Graaff-Reinet. He regularly travels to Adelaide to visit his parents. He also has family that live in Jansenville. The map indicates the names of regional roads in rhombuses and main roads in rectangles. National roads are indicated in a pentagon. The actual kilometre distance between sections of the road is also indicated on the map.

4.1 Use the map above and answer the questions that follow.

4.1.1 Name the regional road Clyde would use to travel from Graaff-Reinet to Adelaide.
(2)

4.1.2 Name THREE towns and /or cities that Clyde would pass on his way to Adelaide.
(3)

4.1.3 Calculate the actual distance between Graaff-Reinet and Jansenville. (3)

4.1.4 In which general direction is Pearston from Jansenville? (2)

4.2 A distance between two points on the map is 3 cm. The actual (real-life) kilometre distance between the two points is 15 km. Determine the scale used on the map in the form 1 :
(4)

[14]

National road Kilometre distance
Regional road
Mathematical Literacy/P1 8
NSC – Grade 11 Exemplar DBE/2013

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QUESTION 5

5.1 Mr Shibambo is the Grade 11 grade head at his school. The school hosts an annual Grade 11 dance. Learners choose the dress code for the event. Mr Shibambo conducts a survey to confirm the dress code for the 2013 event. The results are recorded in the frequency table below.

TABLE 1

DRESS CODE BOYS GIRLS

FORMAL
20 40

TRADITIONAL DRESS
5 15

FANCY DRESS
5 5

CASUAL WEAR
30 10

5.1.1 Calculate the total number of learners who responded to the survey. (2)

5.1.2 Calculate the percentage of the learners who preferred CASUAL wear as the dress code for the Grade 11 dance.
(3)

5.1.3 Which dress code is the least preferred by the Grade 11 learners ? (2)

5.1.4 Use the data in TABLE 1 above to draw a compound bar graph on ANNEXURE A.
(7)

5.1.5 If a Grade 11 learner is chosen at random from the group that was interviewed, what is the probability that the learner :

(a) Preferred formal wear (2)

(b)
Did not prefer fancy dress
(2)

Mathematical Literacy/P1 9
NSC – Grade 11 Exemplar DBE/2013

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5.2 Mr Shibambo recorded the test results for his Grade 11 Mathematical Literacy class in terms of gender.

The results are listed below.

Boys' scores

48 56 57 58 65 66 66

68 75 77 78 81 85 96

Girls' scores

58 75 49 79 39 99 56

67 98 89 59 75 75

5.2.1 Arrange the girls' scores in ascending order. (1)

5.2.2 Write down the girls' modal score. (2)

5.2.3 Calculate the boys' mean score. (3)

5.2.4 Calculate the boys' median score. (2)

5.2.5 Determine the range of the girls' scores. (2)

5.2.6 Write down the probability that a girl chosen at random scored 75 for the test.
(2)

5.2.7 Write down the probability that a boy chosen at random scored more than 75 for the test.
(2)

[32]

TOTAL: 100

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NAME: _____ CLASS: _____

ANNEXUE A

QUESTION 5.1.4

DRESS CODE FOR THE DANCE

051015202530354045

FORMAL

TRADITIONAL

DRESS

FANCY

DRESS

CASUAL

WEAR

Dress codeNumber of learners